

MEB
Midlands Electricity plc

MIDLANDS ELECTRICITY plc

OVERHEAD LINE STANDARDS

STANDARD OHL5
ISSUE 1 - OCTOBER 1992

66kV SINGLE CIRCUIT OVERHEAD LINE
ON WOOD POLES
200 AAAC, 300 AAAC AND 400 ACSE
CONDUCTORS

AUTHORISED BY STANDING INSTRUCTION 10.03

DISTRIBUTION ENGINEERING DIRECTOR

RETAIN2/150/1

I N D E X

<u>CLAUSE</u>		<u>PAGE</u>
1.	Foreword	1
2.	Department of Energy Consents	1
3.	Scope	1
4.	Environmental Conditions	1
5.	Definitions	2
6.	Clearances	2
7.	Engineering Recommendations	2
8.	Design Data	3
9.	Survey	3
10.	Technical Requirements for Erection of Lines	3
11.	Materials	5
12.	Anti Climbing Devices (Normal Arrangements)	6
13.	Anti Climbing Devices (Enhanced Arrangements)	6
14.	Safety Signs	7
15.	Erection of Conductors	7
16.	Index to Tables	8
17.	Index to Drawings	21

66kV SINGLE CIRCUIT OVERHEAD LINES ON WOOD POLES

1. FOREWORD

This standard supersedes the MP14 series 66kV design covered in the Overhead Line Construction Manual.

All new 66kV single circuit lines not requiring uprating to 132kV in the company's long term plan should be designed and constructed in accordance with this standard.

In general, the pole arrangements remain the same and can therefore be used for the replacement of existing poles under refurbishment schemes. Pole arrangements other than those shown in figures 1-6 should only be used with Headquarters approval.

2. DEPARTMENT OF ENERGY CONSENTS

The following reference should be quoted when making application for new lines to the department of Energy:-

Reference:- EL 82/21/7.MEB/66.H.

3. SCOPE

This standard covers 3 phase 66kV lines constructed on wood poles employing the following conductors, designed to operate at 75°C:-

200 sq.mm AAAC code poplar
300 sq.mm AAAC code Upas
400 sq.mm ACSR code Zebra

Lines to this standard are designed to meet the requirements of the 1988 supply regulations and are based on design principles given in Technical Report ET.111.

4. ENVIRONMENTAL CONDITIONS

The two environmental conditions are:-

- a) Normal Environment (site up to 199m AOD) (Above Ordnance Datum)
- b) Moderate Environment (sites between 200-399 AOD)

Requirements for lines operating in severe environments (sites above 400m AOD) are unlikely and have therefore been omitted.

Selection of basic span, span length, support and stay arrangements for the design of lines should be carried out in accordance with Section 10.

Lines that cross environmental boundaries should be designed to the higher standard unless a section pole with fore and aft staying is installed.

5. DEFINITIONS

Intermediate Support: A support in a straight run of line on which conductors are supported by suspension insulators.

Intermediate Angle Support: A support on which a line deviates and the conductors are supported on suspension insulators.

Section Support: A support in a straight run of line on which conductors are made off on either side of the crossarm on Tension Insulator Sets.

Section Angle Support: A support on which a line deviates and conductors are made off on either side of the crossarm on Tension Insulator Sets.

Terminal Support: A support on which conductors are terminated on one side of the crossarm at full line tension on Tension Insulator Sets.

Span: The horizontal distance between adjacent supports.

Wind Loading Span: The wind loading span associated with any support is half the sum of the spans adjacent to the support.

Average Span: The arithmetic average of a number of spans within a section of line.

$$L_{av} = \frac{\sqrt{\frac{L_1^3}{L_1} + \frac{L_2^3}{L_2} + \frac{L_3^3}{L_3} + \text{etc}}}{3}$$

L_{av} = Average Span

$L_1 + L_2$ etc = individual spans within a line or a section of line.

Basic Span: The span length adopted for sag/tension calculations.

Recommended Span: The typical span length in any section of line to which the line should be planned.

Maximum Span: The maximum permitted length of any span.

Sag: The vertical distance under any system of conductor loading, between the conductor and a straight line joining conductor take-off points on adjacent supports, measured out mid span.

6. CLEARANCES

Lines should be designed with a minimum ground clearance of 7m. Clearances to fixed objects should be in accordance with Section 1 of Midlands Electricity Overhead Line Reference Manual.

7. ENGINEERING RECOMMENDATIONS

Engineering Recommendations and Agreements with other bodies relating to Overhead Line Construction and Operation are included in Midlands Electricity Overhead Line Reference Manual.

8. DESIGN DATA

This standard is designated in accordance with the 1988 Electricity Supply regulations on the basis of weather data detailed in Technical Report ET11.

Design Strength of Insulators, fittings, supports (including stays) where fitted shall not have less than 3 times the probable loadings expected over a 50 year return period.

Design strength of conductors shall not be less than 2.5 times the probable loadings expected over a 50 year return period.

Conductor Tension at 15°C in still air shall not exceed 20 per cent of the conductor breaking load.

9. SURVEY

Survey of new lines should be carried out using metric instrumentation and profiles prepared on metric profile paper with a horizontal scale of 1:2000 and vertical scale of 1:200. Metric templates should be used for lines operating under the two environmental conditions.

10. TECHNICAL REQUIREMENTS FOR THE ERECTION OF LINES

10.1 General

Supports shall be erected in accordance with Figures 1-6.

10.2 Maximum and Recommended Span Lengths

Maximum and recommended span lengths are specified in Table 1.

10.3 Wind Loading Spans

Minimum pole sizes for Intermediate and section supports with bare conductors are given in Table 2. Stout class H poles are of sufficient strength for maximum permissible spans for each environmental class.

10.4 Angle of Line Deviation

The maximum angles of line deviation for each support type is listed below:-

Figure 1 0 - 3° 200 AAAC only

Figure 2 3° - 6° 200 AAAC only

Figure 3 0 - 60° Deviation

10.5 Staying Arrangements and Pole Sizes

For Angle, Section Angle and Terminal supports are shown in tables 3 and 4. The recommended stay angle is 45°. The maximum possible stay angle that can be achieved should be used, to reduce the load imposed on poles and stays.

Staying Arrangements and pole loading for lines operating in normal environments (0-199m) are shown on Table 3 staying arrangements and pole loading for lines operating in moderate environment (200-399m AOD) are shown in Table 4.

The vertical stout loads for varying angles of deviation and stay angle shown on tables 3 and 4 should be used for the selection of pole size to withstand applied loads.

The minimum suitable pole diameter for each length of pole can be obtained from table 5.

10.6 Stay Angles and Stay Spread

Stay spreads for the range of permissible stay angles are given in Table 6.

10.7 Stay Separation

The minimum stay separation between adjacent stays must not be less than 1.5m. If stay separation is less than the above figure the uplift of adjacent stays will overlap and reduce the effective stay strength.

10.8 Conductor Design and Erection Tensions

Conductor tensions and sags are given in Table 7a, b and c for Normal Environment (160m basic span) and Table 8a, b and c for moderate environment (120m or 130m basic span).

Erection Tensions include 10% or 12.5% over-tensioning for AAAC and ACSR conductors respectively to allow for conductor settlement.

Where conductors are retensioned 2 years or more after erection, the specified erection tension should be reduced by 10% for AAAC conductors and 12.5% for ACSR conductors.

10.9 Conductor Downpull

Conductors should be sectioned where conductor downpull (measured at pole top) exceeds 1:10 on both sides of pole or 1:5 on one side only. The vertical loads shown on table 3 and 4 should be increased by the following where downpull exists.

Section/Terminal Pole Downpull Loading

Downpull 1:10 (one side)	350 kgf
1:10 (two sides)	700 kgf
1:5 (one side)	700 kgf

11. MATERIALS

Materials specified on opposite page to pole arrangement Drawings (Figs 2-6) are included in the Overhead Line Material Manual.

11.1 Poles

Minimum stout class poles to BS1990. Pole drillings to Fig 1. Unused holes should be plugged with plugs to Drawing M425.

11.2 Conductors

The following conductors are permitted:-

200 sq.mm AAAC (Poplar) stranding 37/2.87mm material etc to BS3242
300 sq.mm AAAC (Upas) to BS3242
400 sq.mm ACSR (Zebra) to BS215 pt 2

Greased to Engineering Recommendation L38/1 Category 3.

11.3 Insulators

Insulators to comply with ESI Standard 43-93.

Suspension Insulators All conductor types 5-70 kN Glass or Porcelain to Drawing M85.

Tension Insulators 200 AAAC conductors 5-70 kN Glass or Porcelain to Drawing M85.

Tension Insulators 300 AAAC and 400 ACSR conductors 5-125 kN Glass or Porcelain to Drawing M254.

Post Insulators All conductor types. Post insulator to Drawing M314. Conductor size to be notified to supplier.

11.4 Stay Arrangements

Stays to ESI Standard 43-91, minimum failing load 110 kN.

Stay Strand 19/3.55mm 700 Grade to BS183 galvanised to BS729.

Stay Block Type 2 wood block to ESI 49-91 to Drawing No M410.

Stay Insulator 130 MFL - Rod type to Drawing No M445.

Pole Top Make Off To be hand wrapped.

Stay Separation Stays to have a minimum separation of 1.5m at ground level.

Stay Compaction

Stay backfilling to be compacted in 200mm layers for 50% of depth then 300 layers for the remainder.

12. ANTI CLIMBING DEVICES (Normal Arrangements)**12.1 Wood Poles**

Poles fitted with cables or earthwires shall be wrapped with twelve turns of barbed wire stapled around the pole, commencing at a minimum of 2.15m from ground level and extending upwards for 0.6m. The barbed wire must be continuous over cables and earth conductors attached to the pole. Suitable precautions must be taken to prevent damage to cables and earth wire insulation

Clean poles in situations listed in 13.1 shall be fitted with 12 turns of barbed wire as above.

12.2 Isolator Operating Rods

Operating rods shall be wrapped separately with barbed wire from a minimum distance from ground level of 2.75m extending upwards for 1.5m with the operating rod in the UP position. Alternatively where an insulating insert is at the lower end of the operating rod, the barbed wire wrap on the rod shall commence immediately above the insert. The barbed wire in this situation shall not encroach on the insulating insert.

12.3 Cable Termination Poles

Outrigger anti-climbing devices type M91 should be used crosslaced between legs and cable support to prevent climbing access between legs.

12.4 Stays

Stays which are in close proximity to each other shall be securely wrapped with barbed wire from a minimum distance from ground of 2.15m extending upwards for a distance of 1.5m. Twin stay insulators shall not be positioned so as to cause shorting or damage due to clashing.

12.5 Barbed Wire

Barbed wire shall be triple life type and shall have a maximum pitch between barbs of 50mm and shall be wrapped to poles and stays with a maximum distance between wraps of 500mm.

13. ANTI-CLIMBING DEVICES (ENHANCED ARRANGEMENTS)

13.1 Enhanced arrangements 13.2 and 13.3 shall be used where there is a higher probability of unauthorised climbing such as:-

- (a) Schools or official recreation and play areas.

- (b) Known unauthorised play areas.
- (c) Residential and non-residential caravan/chalet parks.
- (d) Gypsy encampments.
- (e) Fishing areas, sailing waters, picnic areas, parks and sports grounds.
- (f) Areas of high vandalism and crime.
- (g) Supports situated in places adjacent to areas (a) to (f) above.
- (h) Nearness to trees, fences, buildings, walls and particularly if trees are easily climbable.

Consultation should take place with landowners and other interested parties to enable the correct decision to be made on types of anti climbing devices used.

13.2 Stays

Single stays with a stay angle exceeding 45° shall be fitted with barbed wire as 12.4 and 12.5.

13.3 H Poles with or without Isolator Operating Rods

Outrigger anti-climbing devices type M91 should be used, crosslaced between legs to prevent climbing.

14. SAFETY SIGNS

Two safety signs to be fitted, one on each leg on opposite faces.

15. ERECTION OF CONDUCTORS

Conductor should only be erected by staff who have received instruction in the use of materials specified.

Winches, cable drums and equipment should only be used by staff who have received training in their use.

The correct use should be made of scaffolding or approved temporary supports during conductor erection for the protection of the general public and property.

The requirements of Engineering Recommendation G.35 should be met during construction work.

Landowners, tenants and other parties who could be affected during conductor erection should be advised of the extent of work involved and given notice of when work is to be carried out.

16. INDEX TO TABLES

<u>TABLE</u>	<u>PAGE</u>
1. Recommended Span Lengths	9
2. Maximum Span Lengths of Stout Poles	10
3. Pole Loading/Stay Requirements (0-199m) AOD	11
4. Pole Loading/Stay Requirements (200-399mm) AOD	12
5. Maximum Permissible Strut Pole Loading	13
6. Stay Angles and Stay Spread	14
7a. Erection Table 200 sq.mm AAAC (0-199m AOD)	15
7b. Erection Table 300 sq.mm AAAC (0-199m AOD)	16
7c. Erection Table 400 sq.mm ACSR (0-199m AOD)	17
8a. Erection Table 200 sq.mm AAAC (200m-399m AOD)	18
8b. Erection Table 300 sq.mm AAAC (200m-399m AOD)	19
8c. Erection Table 400 sq.mm ACSR (200m-399m AOD)	20

RECOMMENDED SPAN LENGTH

SITE AOD (m)	CONDUCTOR TYPE AND SIZE	BASIC SPAN (m)	RECOMMENDED SPAN (m)	SPAN RANGE (m)
0-199m	200 sq mm AAAC	160	150-160	120-180
	300 sq mm AAAC	160	150-160	120-180
	400 sq mm ACSR	160	150-160	120-180
200+	200 sq mm AAAC	130	120-130	100-150
	300 sq mm AAAC	130	120-130	100-150
	400 sq mm ACSR	120	120-130	90-140

MAXIMUM SPAN LENGTHS FOR STOUT CLASS POLES
'H' INTERMEDIATE AND STRAIGHT LINE SECTION SUPPORTS

CONDUCTOR TYPE & SIZE	MAXIMUM SPAN LENGTH (m)	
	0-199m AOD	200-399m AOD
200 AAAC	180	150
300 AAAC	180	150
400 ACSR	180	140

Note: Stout class poles meet maximum span requirements of Table 1.

STANDARD STOUT CLASS POLE DIMENSIONS

<u>POLE LENGTH</u>	<u>PLANING DEPTH (m)</u>	<u>DIAMETER (AT 1.5m FROM BUTT)</u>	<u>MIN</u>
10.0m	1.8	285	3443100
10.5m	1.8	290	3443105
11.0m	2.1	295	3443110
11.5m	2.1	300	3443115
12.0m	2.1	305	3443120
13.0m	2.1	320	3443130
14.0m	2.1	335	3443140
15.0m	2.4	350	3443150
16.0m	2.4	365	3443160
17.0m	2.4	375	3443170
18.0m	2.4	390	3443180

**POLE LOADING/STAYING REQUIREMENTS ALL CONDUCTOR TYPES
NORMAL ENVIRONMENT (0-199m ACD)**

ANGLE OF DEVIATION	VERTICAL STRUT LOADING ON STAYED POLES (Kgf) AND STAY REQUIREMENT				
	30 DEG STAY	35 DEG STAY	40 DEG STAY	45 DEG STAY	50 DEG STAY
0-6 DEG	1-2670	1-2440	1-2260	1-2120	1-2000
7-10 DEG	1-3460	1-3090	1-2800	1-2670	1-2380
11-15 DEG	2-4440	1-3900	1-3480	1-3140	1-2850
16-20 DEG	2-5410	2-4700	2-4150	1-3700	1-3330
21-25 DEG	2-6380	2-5500	2-4820	2-4550	2-3790
26-30 DEG	3-4890	2-6290 3-4200	2-5480 3-3650	2-4810 3-3210	2-4260 3-2840
31-35 DEG	3-5530 4-4610	3-4720 4-3930	3-4090 4-3410	3-3580 4-2980	3-3150 4-2620
36-40 DEG	4-5120	3-5230 4-4360	3-4520 4-3760	3-3940 4-3280	3-3450 4-2880
41-45 DEG	4-5640	4-4780	3-4940 4-4120	3-4290 4-3580	3-3750 4-3120
46-50 DEG	4-6140	4-5190	4-4460	4-3870	4-3370
51-55 DEG	4-6630	4-5600	4-4800	4-4150	4-3600
56-60 DEG	4-7110	4-6000	4-5130	4-4430	4-3840
TERMINAL	4-6860	4-5740	4-4880	4-4170	4-3580

- NOTES 1) All stays 19/3.55.
 2) Pole loadings (Kgf) are for max. loaded leg.
 3) Recommended stay angle 45°.
 4) Max. permissible loading of suitable poles given in table 5.

**POLE LOADING/STAYING REQUIREMENTS ALL INSULATOR TYPES
MODERATE/SEVERE ENVIRONMENT (200-399 AOD)**

ANGLE OF DEVIATION	VERTICAL STRUT LOADING ON STAYED POLES (Kgf) AND STAY REQUIREMENT				
	30 DEG STAY	35 DEG STAY	40 DEG STAY	45 DEG STAY	50 DEG STAY
0-6 DEG	1-2970	1-2710	1-2510	1-2350	1-2210
7-10 DEG	1-3870	1-3460	1-3130	1-2870	1-2650
11-15 DEG	2-4990	2-4380	1-3900	1-3520	1-3190
16-20 DEG	2-6100	2-5300	2-4670	2-4160	2-3730
21-25 DEG	2-7210	2-6210	2-5430	2-4800	2-4270
26-30 DEG	3-5540	3-4740	2-6180 3-4120	2-5430 3-3620	2-4800 3-3200
31-35 DEG	4-5210	3-5330 4-4450	3-4620 4-3850	3-4040 4-3360	3-3550 4-2960
36-40 DEG	4-5810	4-4930	3-5110 4-4260	3-4450 4-3710	3-3890 4-3240
41-45 DEG	4-6390	4-5420	4-4660	3-4850 4-4040	3-4230 4-3530
46-50 DEG	4-6960	4-5890	4-5050	4-4370	4-3800
51-55 DEG	N/A	4-6350	4-5440	4-4700	4-4080
56-60 DEG	N/A	4-6800	4-5810	4-5010	4-4340
TERMINAL	N/A	4-6520	4-5540	4-4740	4-4060

- NOTES 1) All stays 19/3.55.
 2) Pole loadings (Kgf) are for max. loaded leg.
 3) Recommended stay angle 45°.
 4) Max. permissible loading of suitable poles given in table 5.

MAXIMUM PERMISSIBLE STRUT LOADS FOR STAYED POLES

MAXIMUM LOAD (Kgf)	POLE LENGTH (M) AND DIAMETERS AT 1.7M (MM)										
	10	10.5	11	11.5	12	13	14	15	16	17	18
2600 3100											
2800 3300											390
3000 3600										375	390
3200 3800								350	365	380	395
3400 4100							335	353	370	385	400
3600 4300						320	340	360	375	390	405
3800 4600					305	325	345	360	380	390	410
4000 4800				300	310	325	345	365	380	395	410
4200 5000				305	310	330	350	370	385	400	415
4400 5300		290	295	305	315	335	355	370	390	400	420
4600 5500		295	300	310	315	335	355	375	390	405	420
4800 5800	285	295	300	310	320	340	360	380	395	410	425
5000 6000	290	300	305	315	325	340	360	380	395	410	430
5200 6200	290	300	305	315	325	345	365	385	400	415	430
5400 6500	295	305	310	320	330	345	370	385	405	415	435
5600 6700	295	305	310	320	330	350	370	390	405	420	445
5800 6900	300	310	315	325	335	350	375	390	410	420	455
6000 7200	300	310	315	325	335	355	375	395	410	430	
6200 7400	300	310	320	330	335	355	380	395	415	440	
6400 7700	305	315	320	330	340	360	380	400	415		
6600 7900	305	315	320	330	340	360	385	400	420		
6800 8100	305	315	325	335	345	365	385	405	425		
7000 8400	310	320	325	335	345	365	400	405			
7200 8600	310	320	325	335	345	365	400	410			

- NOTES: 1) POLE DIAMETERS SHOWN ARE STOUT CLASS RANGE
 2) MIN DIAMETERS QUOTED FOR EACH POLE LENGTH ARE MINIMUM BS 1950 DIAMETERS AT 1.5m FROM BUTT
 3) MAXIMUM PERMISSIBLE STRUT LOADS QUOTED ARE FOR TOP DIAMETERS NOT LESS THAN 200mm

STAY ANGLES AND STAY SPREAD

POLE DETAILS			STAY ANGLES AND STAY SPREAD				
POLE LENGTH (m)	PLANTING DEPTH (m)	HEIGHT ABOVE G.L. (m)	30 DEG ANGLE	35 DEG ANGLE	40 DEG ANGLE	45 DEG ANGLE	50 DEG ANGLE
10	1.8	8.2	4.7 (5.6)	5.7 (6.8)	6.9 (8.1)	8.2 (9.7)	9.7 (11.5)
10.5	1.8	8.7	5.0 (5.9)	6.0 (7.1)	7.3 (8.5)	8.7 (10.2)	10.3 (12.1)
11	2.1	8.9	5.1 (6.0)	6.2 (7.3)	7.4 (8.6)	8.9 (10.4)	10.6 (12.4)
11.5	2.1	9.4	5.4 (6.3)	6.5 (7.6)	7.8 (9.0)	9.4 (10.9)	11.2 (13.0)
12	2.1	9.9	5.7 (6.6)	6.9 (8.0)	8.3 (9.5)	9.9 (11.4)	11.7 (13.5)
13	2.1	10.9	6.3 (7.2)	7.6 (8.7)	9.1 (10.3)	10.9 (12.4)	12.9 (14.7)
14	2.1	11.9	6.8 (7.7)	8.3 (9.4)	10.0 (11.2)	11.9 (13.4)	14.1 (15.9)
15	2.4	12.6	7.2 (8.1)	8.8 (9.9)	10.5 (11.7)	12.6 (14.1)	15.0 (16.8)
16	2.4	13.6	7.5 (8.4)	9.5 (10.6)	11.4 (12.6)	13.6 (15.1)	16.2 (18.0)
17	2.4	14.6	8.4 (9.3)	10.2 (11.3)	12.2 (13.4)	14.6 (16.1)	17.3 (19.1)
18	2.4	15.6	9.0 (9.9)	10.9 (12.0)	13.0 (14.2)	15.6 (17.1)	18.5 (20.3)

NOTE (1) STAY SPREAD SHOWN ARE AT GROUND LEVEL

(2) FIGURES IN BRACKETS ARE STAY SPREADS AT PLANTING DEPTH
ie Distance from pole to front edge of excavation or stay block.

ERECTION TABLE - TABLE - 200 SQ MM AAAC 160M BASIC SPAN
NORMAL ENVIRONMENT (0 - 199m AOD)
CONDUCTOR TENSIONS INCLUDE 10% OVER-TENSIONING

TEMP °C	TENSION kgf	SAGS FOR ACTUAL SPAN (M)						
		120	130	140	150	160	170	180
-10	1594	0.77	0.90	1.05	1.20	1.37	1.54	1.73
- 8	1547	0.79	0.93	1.08	1.24	1.41	1.59	1.78
- 6	1501	0.82	0.96	1.11	1.28	1.45	1.64	1.84
- 4	1457	0.84	0.99	1.14	1.31	1.50	1.69	1.89
- 2	1414	0.87	1.02	1.18	1.35	1.54	1.74	1.95
0	1372	0.89	1.05	1.22	1.40	1.59	1.79	2.01
2	1332	0.92	1.08	1.25	1.44	1.64	1.85	2.07
4	1294	0.95	1.11	1.29	1.48	1.68	1.90	2.13
6	1257	0.98	1.14	1.33	1.52	1.73	1.96	2.19
8	1221	1.00	1.18	1.37	1.57	1.78	2.01	2.26
10	1187	1.03	1.21	1.40	1.61	1.84	2.07	2.32
12	1154	1.06	1.25	1.44	1.66	1.89	2.13	2.39
14	1123	1.09	1.28	1.48	1.70	1.94	2.19	2.45
16	1093	1.12	1.32	1.53	1.75	1.99	2.25	2.52
18	1065	1.15	1.35	1.57	1.80	2.05	2.31	2.59
20	1038	1.18	1.39	1.61	1.84	2.10	2.37	2.66
22	1012	1.21	1.42	1.65	1.89	2.15	2.43	2.72
24	987	1.24	1.46	1.69	1.94	2.21	2.49	2.79
26	964	1.27	1.49	1.73	1.99	2.26	2.55	2.86
28	942	1.30	1.53	1.77	2.03	2.31	2.61	2.93
30	920	1.33	1.57	1.81	2.08	2.36	2.67	3.00

ERECTION TABLE - 300 SQ.MM AAAC 160m BASIC SPAN
 NORMAL ENVIRONMENT (0-199m AOD)
 CONDUCTOR TENSIONS INCLUDE 10% OVER-TENSIONING

TEMP °C	TENSION kgf	SAGS FOR ACTUAL SPAN (M)						
		120	130	140	150	160	170	180
-10	1326	1.40	1.64	1.90	2.18	2.48	2.80	3.14
- 8	1298	1.43	1.68	1.94	2.23	2.54	2.86	3.21
- 6	1272	1.43	1.71	1.98	2.29	2.59	2.92	3.28
- 4	1246	1.49	1.74	2.02	2.32	2.64	2.98	3.35
- 2	1222	1.52	1.78	2.06	2.37	2.70	3.04	3.41
0	1199	1.55	1.81	2.10	2.41	2.75	3.10	3.48
2	1177	1.57	1.85	2.14	2.46	2.80	3.16	3.54
4	1155	1.60	1.88	2.18	2.51	2.85	3.22	3.61
6	1135	1.63	1.92	2.22	2.55	2.90	3.28	3.67
8	1116	1.66	1.95	2.26	2.60	2.95	3.33	3.74
10	1097	1.69	1.98	2.30	2.64	3.00	3.39	3.80
12	1079	1.72	2.02	2.34	2.68	3.05	3.45	3.86
14	1062	1.75	2.05	2.38	2.73	3.10	3.50	3.93
16	1045	1.77	2.08	2.41	2.77	3.15	3.56	3.99
18	1030	1.80	2.11	2.45	2.81	3.20	3.61	4.05
20	1014	1.83	2.15	2.49	2.86	3.25	3.67	4.11
22	1000	1.85	2.18	2.52	2.90	3.30	3.72	4.17
24	985	1.88	2.21	2.56	2.94	3.34	3.78	4.23
26	972	1.91	2.24	2.60	2.98	3.39	3.83	4.29
28	959	1.93	2.27	2.63	3.02	3.44	3.88	4.35
30	946	1.96	2.30	2.67	3.06	3.49	3.93	4.41

ERECTION TABLE - 400 SQ.MM ACSR 160m BASIC SPAN
NORMAL ENVIRONMENT (0-199m AOD)
CONDUCTOR TENSIONS INCLUDE 12.5% OVER-TENSIONING

TEMP °C	TENSION kgf	SAGS FOR ACTUAL SPAN (M)						
		120	130	140	150	160	170	180
-10	1271	2.37	2.79	3.23	3.71	4.22	4.77	5.34
- 8	1259	2.40	2.81	3.26	3.74	4.26	4.81	5.39
- 6	1247	2.42	2.84	3.29	3.78	4.30	4.85	5.44
- 4	1236	2.44	2.86	3.32	3.81	4.34	4.90	5.49
- 2	1255	2.46	2.89	3.35	3.85	4.38	4.94	5.54
0	1215	2.48	2.92	3.38	3.88	4.42	4.99	5.59
2	1204	2.51	2.94	3.41	3.91	4.45	5.03	5.64
4	1194	2.53	2.97	3.44	3.95	4.49	5.07	5.69
6	1184	2.55	2.99	3.47	3.98	4.53	5.12	5.73
8	1174	2.57	3.02	3.50	4.02	4.57	5.16	5.78
10	1164	2.59	3.04	3.53	4.05	4.61	5.20	5.83
12	1155	2.61	3.07	3.56	4.08	4.64	5.24	5.88
14	1146	2.63	3.09	3.58	4.11	4.68	5.28	5.92
16	1137	2.65	3.11	3.61	4.15	4.72	5.32	5.97
18	1129	2.67	3.14	3.64	4.18	4.75	5.37	6.02
20	1120	2.69	3.16	3.67	4.21	4.79	5.41	6.06
22	1111	2.71	3.19	3.69	4.24	4.83	5.45	6.11
24	1103	2.74	3.21	3.72	4.27	4.86	5.49	6.15
26	1095	2.75	3.23	3.75	4.30	4.90	5.53	6.20
28	1087	2.77	3.26	3.78	4.34	4.93	5.57	6.24
30	1079	2.79	3.28	3.80	4.37	4.97	5.61	6.28

ERECTION TABLE - 200 SQ MM AAC 130M BASIC SPAN
MODERATE ENVIRONMENT (200m - 399m AOD)
CONDUCTOR TENSIONS INCLUDE 10% OVER-TENSIONING

TEMP °C	TENSION kgf	SAGS FOR ACTUAL SPAN (M)						
		90	100	110	120	130	140	150
-10	1886	0.37	0.45	0.55	0.65	0.76	0.88	1.02
- 8	1827	0.38	0.47	0.56	0.67	0.79	0.91	1.05
- 6	1770	0.39	0.48	0.58	0.69	0.81	0.94	1.08
- 4	1713	0.40	0.50	0.60	0.72	0.84	0.97	1.12
- 2	1658	0.42	0.51	0.62	0.74	0.87	1.01	1.15
0	1604	0.43	0.53	0.64	0.76	0.90	1.04	1.19
2	1550	0.44	0.55	0.66	0.79	0.93	1.08	1.23
4	1499	0.46	0.57	0.69	0.82	0.96	1.11	1.28
6	1448	0.48	0.59	0.71	0.85	0.99	1.15	1.32
8	1399	0.49	0.61	0.74	0.88	1.03	1.19	1.37
10	1352	0.51	0.63	0.76	0.91	1.06	1.23	1.42
12	1306	0.53	0.65	0.79	0.94	1.10	1.28	1.47
14	1262	0.55	0.67	0.82	0.97	1.14	1.32	1.52
16	1219	0.57	0.70	0.84	1.01	1.18	1.37	1.57
18	1178	0.59	0.72	0.87	1.04	1.22	1.42	1.63
20	1139	0.61	0.75	0.90	1.08	1.26	1.46	1.68
22	1102	0.63	0.77	0.93	1.11	1.31	1.51	1.74
24	1066	0.65	0.80	0.97	1.15	1.35	1.56	1.80
26	1032	0.67	0.82	1.00	1.19	1.39	1.62	1.86
28	999	0.69	0.85	1.03	1.23	1.44	1.67	1.92
30	966	0.71	0.89	1.07	1.27	1.50	1.73	1.99

ERECTION TABLE - 300 sq mm AAAC 130m BASIC SPAN
MODERATE ENVIRONMENT (200m - 399m AOD)
CONDUCTOR TENSIONS INCLUDE 10% OVER-TENSIONING

TEMP °C	TENSION kgf	SAGS FOR ACTUAL SPAN (M)						
		90	100	110	120	130	140	150
-10	1521	0.69	0.85	1.02	1.22	1.43	1.66	1.90
- 8	1474	0.71	0.87	1.06	1.26	1.48	1.71	1.96
- 6	1430	0.73	0.90	1.09	1.30	1.52	1.76	2.03
- 4	1387	0.75	0.93	1.12	1.34	1.57	1.82	2.09
- 2	1347	0.77	0.96	1.16	1.38	1.61	1.87	2.15
0	1309	0.80	0.98	1.19	1.42	1.66	1.93	2.21
2	1273	0.82	1.01	1.22	1.46	1.71	1.98	2.27
4	1239	0.84	1.04	1.26	1.50	1.76	2.04	2.34
6	1207	0.86	1.07	1.29	1.53	1.80	2.09	2.40
8	1177	0.89	1.09	1.31	1.57	1.85	2.14	2.46
10	1147	0.91	1.12	1.36	1.62	1.90	2.20	2.52
12	1120	0.93	1.15	1.39	1.65	1.94	2.25	2.59
14	1094	0.95	1.18	1.42	1.69	1.99	2.31	2.65
16	1069	0.97	1.20	1.46	1.73	2.03	2.36	2.71
18	1046	1.00	1.23	1.49	1.77	2.08	2.41	2.77
20	1024	1.02	1.26	1.52	1.81	2.12	2.46	2.83
22	1003	1.04	1.28	1.55	1.85	2.17	2.51	2.89
24	982	1.06	1.31	1.58	1.89	2.21	2.57	2.95
26	963	1.08	1.34	1.62	1.92	2.26	2.62	3.01
28	945	1.10	1.36	1.65	1.96	2.30	2.67	3.06
30	927	1.12	1.39	1.68	2.00	2.34	2.72	3.12

ERECTION TABLE - 400 SQ MM ACSR 120M BASIC SPAN
MODERATE ENVIRONMENT (200m - 399m AOD)
CONDUCTOR TENSIONS INCLUDE 12.5% OVER-TENSIONING

TEMP °C	TENSION kgf	SAGS FOR ACTUAL SPAN (M)						
		90	100	110	120	130	140	150
-10	1535	1.11	1.36	1.65	1.97	2.31	2.67	3.07
- 8	1509	1.12	1.39	1.68	2.00	2.35	2.72	3.12
- 6	1484	1.14	1.41	1.71	2.03	2.39	2.77	3.18
- 4	1460	1.16	1.44	1.74	2.07	2.43	2.81	3.23
- 2	1436	1.18	1.46	1.77	2.10	2.47	2.86	3.28
0	1414	1.20	1.48	1.79	2.13	2.50	2.90	3.33
2	1393	1.22	1.50	1.82	2.17	2.54	2.95	3.38
4	1373	1.24	1.53	1.85	2.20	2.58	2.99	3.43
6	1353	1.25	1.55	1.87	2.23	2.62	3.04	3.48
8	1333	1.27	1.57	1.90	2.26	2.66	3.08	3.54
10	1315	1.29	1.59	1.93	2.29	2.69	3.12	3.59
12	1298	1.31	1.61	1.95	2.33	2.73	3.16	3.63
14	1280	1.33	1.64	1.98	2.36	2.77	3.21	3.68
16	1264	1.34	1.66	2.01	2.39	2.80	3.25	3.73
18	1248	1.36	1.68	2.03	2.42	2.84	3.29	7.78
20	1232	1.38	1.70	2.06	2.45	2.87	3.33	3.83
22	1217	1.39	1.72	2.08	2.48	2.91	3.37	3.87
24	1203	1.41	1.74	2.11	2.51	2.94	3.41	3.92
26	1189	1.43	1.76	2.13	2.54	2.98	3.45	3.96
28	1175	1.44	1.78	2.16	2.57	3.01	3.49	4.01
30	1161	1.46	1.80	2.19	2.60	3.05	3.53	4.06

17. INDEX TO DRAWINGS

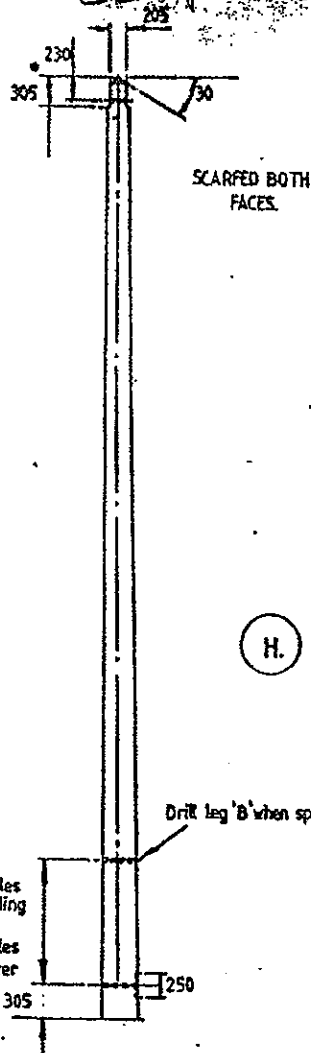
FIGURE

1. Pole Drilling Details
2. Straight Line Suspension Support
3. Angle Suspension Support
4. Section Angle Support
5. Terminal Support
6. Terminal Support XLP U/G Cable
7. Suspension Insulator Assembly 200 AAAC
8. Suspension Insulator Assembly 300 AAAC
9. Suspension Insulator Assembly 400 ACSR
10. Tension Insulator Assembly 200 AAAC
11. Tension Insulator Assembly 300 AAAC
12. Tension Insulator Assembly 400 ACSR
13. Fandown Arrangement All Conductors

NOTE: ALL POLE DRILLINGS 22mm

225 for terminal

EDS



H.

1-219 For poles
upto & including
14-00m.
1-524 For poles
14-00m & over

H MP 11/3
H MP 11/4
H MP 11/4 HS.
H MP 11/5
H MP 11/6
H MP 11/6 HS
H MP 13/3
H MP 13/5
H MP 14/1
H MP 14/2
H MP 14/3
H MP 14/4

FOUNDATION BLOCKS IF REQUIRED.

MP 11 & 13 SERIES

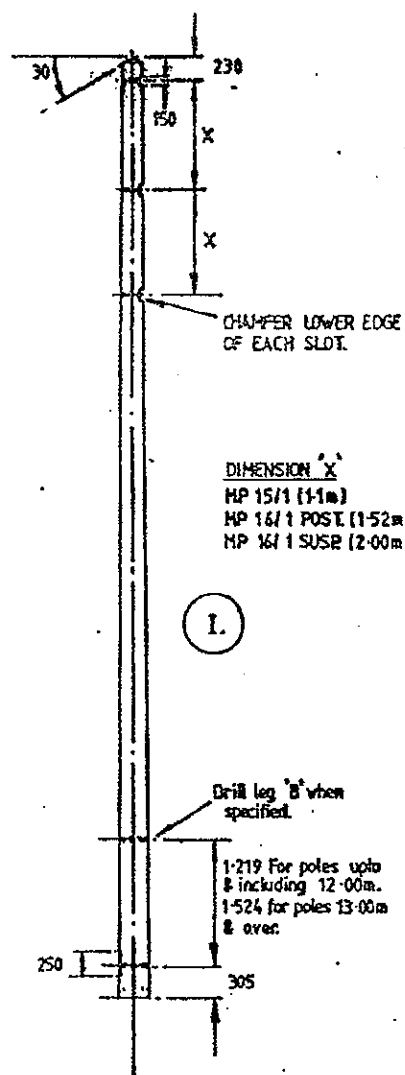
2-M 441/1
1-M 441/3
1-M 441/4 or 5

MP 14/1 & 14/2

2-M 441/1
1-M 442/1
1-M 442/2 or 3

MP 14/3 & 14/4

2-M 441/1
1-M 442/4
1-M 442/5 or 6



I.

DIMENSION 'X'
MP 15/1 (1.1m)
MP 16/1 POST (1.52m)
MP 16/1 SUSP (2.00m)

Drill leg 'B' when
specified.

1-219 For poles upto
& including 12-00m.
1-524 for poles 13-00m
& over.

H MP 15/1
H MP 16/1 POST
H MP 16/1 SUSP

FOUNDATION BLOCKS IF REQUIRED.

MP 15/1

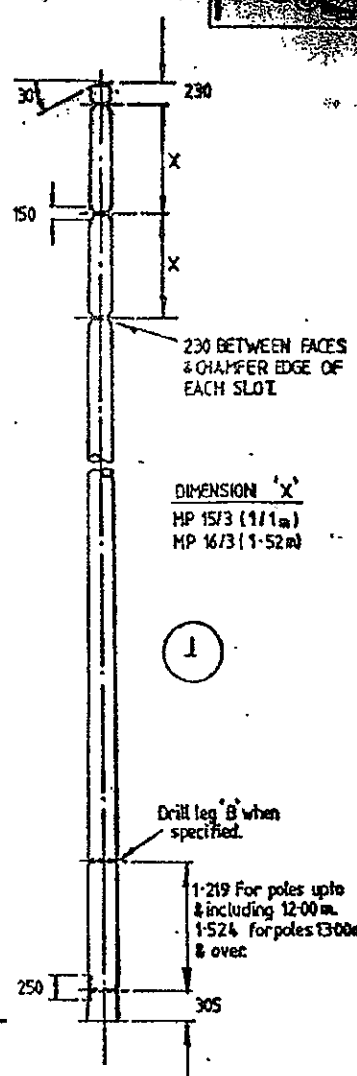
2-M 441/1
1-M 442/4
1-M 442/5 or 6

MP 16/1 POST

2-M 441/1
1-M 442/1
1-M 442/2 or 3

MP 16/1 SUSP

2-M 441/1
1-M 442/4
1-M 442/5 or 6



J.

DIMENSION 'X'
MP 15/3 (1.1m)
MP 16/3 (1.52m)

Drill leg 'B' when
specified.

1-219 For poles upto
& including 12-00m.
1-524 for poles 13-00m
& over.

H MP 15/3
H MP 16/3

FOUNDATION BLOCKS IF REQUIRED.

2-M 441/1
1-M 442/4
1-M 442/5 or 6

NOTE: TREATED AS E.S.I. STD. 63-88 ISSUE 2.

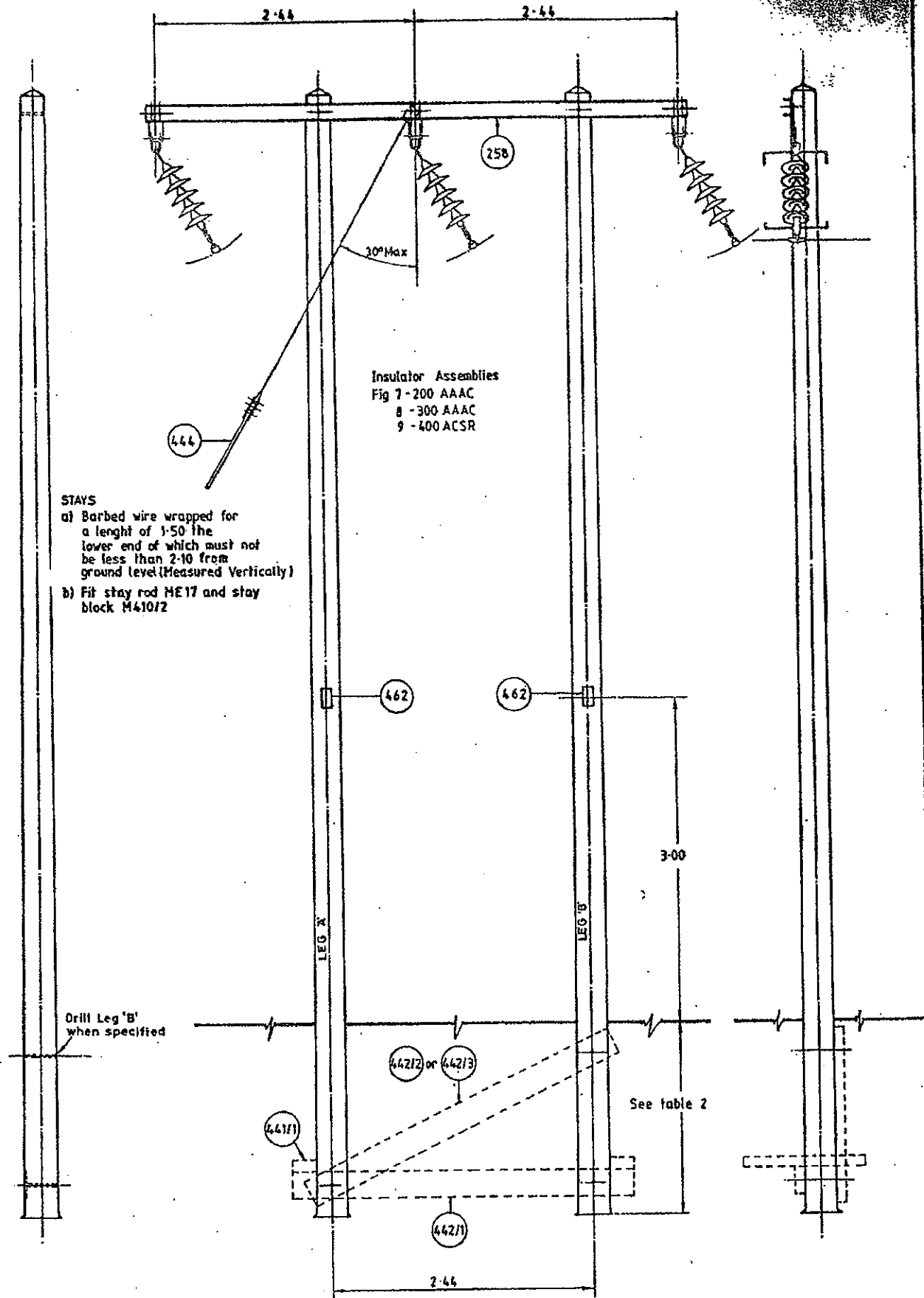
AS DRG. M450/4

DRAWN BY		TITLE POLE DRILLINGS MP TYPES AS SHOWN	SCALE DATE 27-5-92 DRAWN R.A.C. TRACE CHECKED APPROVED	Midlands Electricity plc FIG. 1
----------	--	--	---	--

MATERIAL SCHEDULE FIG 2 (HP14/1)

ITEM	MIN	QUANTITY
Pole (size to be specified - see table 2)	3453—	2
Crossarm 200 AAAC Conductors	3451000	)
300 AAAC Conductors)		1
400 AAAC Conductors)	3452580	)
'U' link	3451041	3
Ball Ended Eye Link 70KN MFL	3451330	3
Disc Insulator Porcelain 70KN MFL	3450851	)
Glass 70KN MFL	3450852	) 15
Socket Tongue 70KN MFL	3450880	3
Suspension Clamp 300 AAAC Conductor	3454931	)
300 AAAC Conductor	3454932	) 3
400 ACSR Conductor	3454933	)
Arc Horn Live End	3450891	3
Dead End	3450892	3
Safety Sign Divs 3, 5 and 8	3454621	)
Divs 1, 4 and 7	3454622	) 2
Foundation Blocks (if required)		
1.25m x 250 x 100	3454411	2
3.5m x 250 x 100	3454421	1
3.23m x 250 x 100 Poles up to 14m	3454422	)
3.38m x 250 x 100 Poles over 14m	3454423	) 1
Assembly Bolts M20 x 45	3453661	6
Pole Bolt - Crossarm - M20 x 260	3453633	2
Foundation Bolts (if required)		
M20 x 450	3453636	)
M20 x 530	3453637	)
M20 x 610	3453638	) 5
M20 x 760	3453639	)
Square Flat Washer - M20 x 65	3453170	2-12*

* With Foundation Blocks



For Pole fabrication
see Fig 1(H)

AS ORG MP14/2

TITLE

66kV CONSTRUCTION
'H' SUSPENSION THRO' ANGLES

SCALE	
DATE	27.5.92
DRAWN	F. A. H.
TRACED	L. F. C.
CHECKED	
APPROVED	

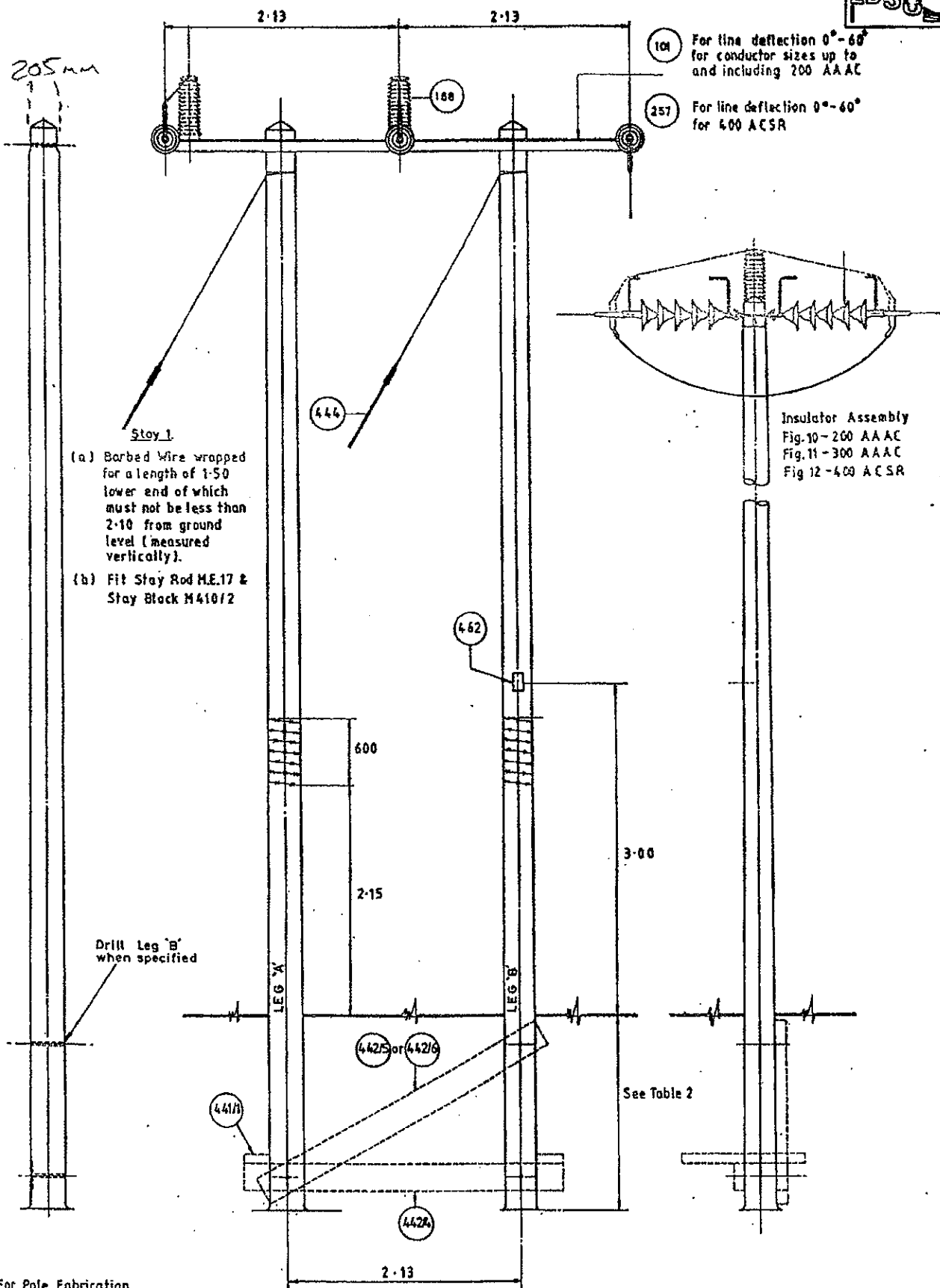
MEB
Midlands Electricity plc

FIG. 3.

MATERIAL SCHEDULE FIG 3 (NP14/2)

ITEM	MIN	QUANTITY
Pole (size to be specified)	SPECIAL	2
Crossarm all Conductors	3452580	1
Gusset Plate	3451050	3
'U' Link	3451000	3
Ball Ended Eye Link 70KN MFL	3451330	3
Disc Insulator Porcelain 70KN MFL	3450851	)
Glass 70KN MFL	3450852	) 15
Socket Tongue 70KN MFL	3450880	3
Suspension Clamp 200 AAAC Conductors	3454931	)
300 AAAC Conductors	3454932	) 3
400 ACSR Conductors	3454933	)
Arc Horn Live End	3450891	3
Dead End	3450892	3
Safety Sign Dirs 3, 5 and 8	3454621	)
Dirs 1, 4 and 7	3454622	) 2
Foundation Blocks (if required)		
1.25m x 250 x 100	3454412	2
3.5m x 250 x 100	3454421	1
3.23m x 250 x 100 Poles up to 14m	3454422	)
3.38m x 250 x 100 Poles over 14m	3454423	) 1
Assembly Bolts M20 x 45	3452661	12
Pole Bolt - Crossarm - M20 x 26	3453633	2
Foundation Bolts (if required)		
M20 x 450	3453636	)
M20 x 530	3453637	)
M20 x 610	3453638	) 5
M20 x 760	3453639	)
Square Flat Washer - M20 x 65	3453170	2-12*
Stay Terminating Strap	3451460	1
Stay Wire 19/3.55-700 Grade	3454562	AR
Dead End Stay Grip	3452703	4
Stay Insulator	3454450	1
Stay Rod	3450172	1
Stay Block	3454102	1
Stay Thimble	3451152	4

* With Foundation Blocks



AS DRG MP 14/3

TITLE

66 kV. CONSTRUCTION
'H' SECTION POLE

SCALE	27.5:92
DATE	F.A.H.
DRAWN	J.P.
TRACED	
CHECKED	
APPROVED	

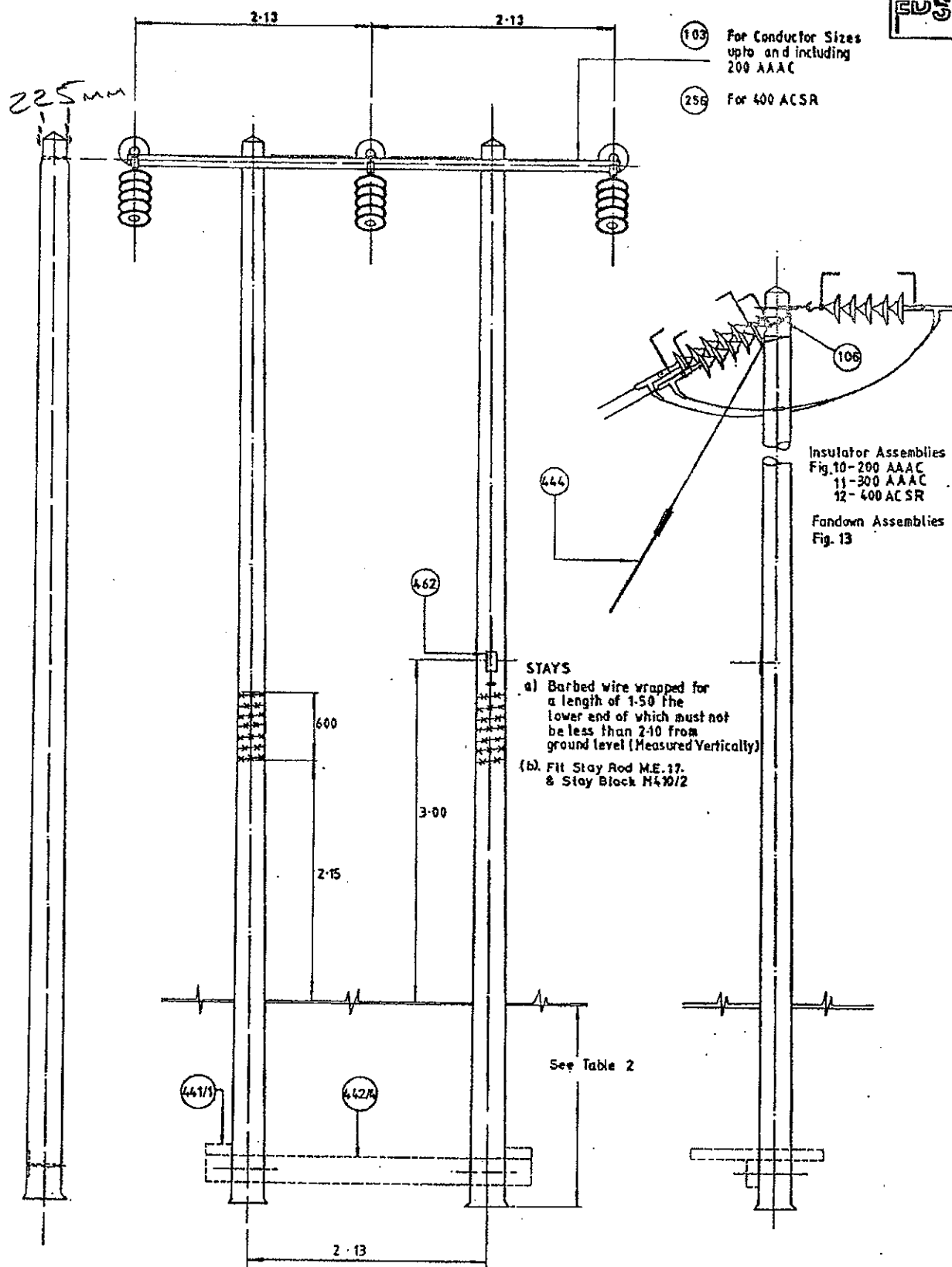
MEB

Midlands Electricity plc

FIG. 4.

MATERIAL SCHEDULE FIG 4 (NP14/3)

ITEM	MIN	QUANTITY
Pole (size to be specified)	SPECIAL	2
Crossarm 200 AAAC Conductors	3451010	) 1
300 AAAC Conductors)		) or
400 ACSR Conductors)	3452570	1
<u>Insulator Assembly 200 AAAC</u>		
Ball Ended Hook 70KN MFL	3450860	6
Disc Insulator Porcelain 70KN MFL	3450851	)
Glass 70KN MFL	3450852	) 30
Socket Tongue 70KN MFL	3450880	6
Arc Horn Live End	3450893	6
Dead End	3450894	6
<u>Insulator Assembly 300 AAAC + 400 ACSR</u>		
Shackle 125KN MFL	3452500	6
Ball Ended Eye Link 125KN MFL	3452490	6
Disc Insulator Porcelain 125KN MFL	3452541	30
300 AAAC Socket Tongue 125KN MFL	3452511	6
or 400 ACSR Socket Clevis 125KN MFL	3455100	6
+ Adaptor 125KN MFL	3453601	6
Arc Horn Live End	3452552	6
Dead End	3452551	6
Conductor Terminating Clamp 200 AAAC 70KN	3455681	)
300 AAAC 125KN	3455684	) 6
400 ACSR	3452531	)
Post Insulator 300 AAAC + 400 ACSR	3453140	) 2
Helical Tie 200 AAAC only	3454768	2
Safety Sign Divs 3, 5 and 8	3454621	)
Divs 1, 4 and 7	3454622	) 2
Foundation Blocks (if required)		
1.25m x 250 x 100	3454411	2
3.2m x 250 x 100	3454424	1
2.96m x 250 x 100 Poles up to 14m	3454425	)
or 3.12m x 250 x 100 Poles over 14m	3454426	) 1
Pole Bolt - Crossarm - M20 x 260	3453633	2
Foundation Bolts (if required)		
M20 x 450	3453636	)
M20 x 530	3453637	)
M20 x 610	3453638	) 5
M20 x 760	3453639	)
Square Flat Washer - M20 x 65	3453170	0-12*
Stays - Material/Stay		
Stay Wire - 19/3.55 - 700 Grade	3454562	AR
Stay Insulator	3454450	1
Dead End Stay Grip	3452703	3
Stay Thimble	3451152	3
Stay Rod	3450172	1
Stay Block	3454102	1
* With Foundation Blocks		



For Pole Fabrication
See Fig. 1(H)

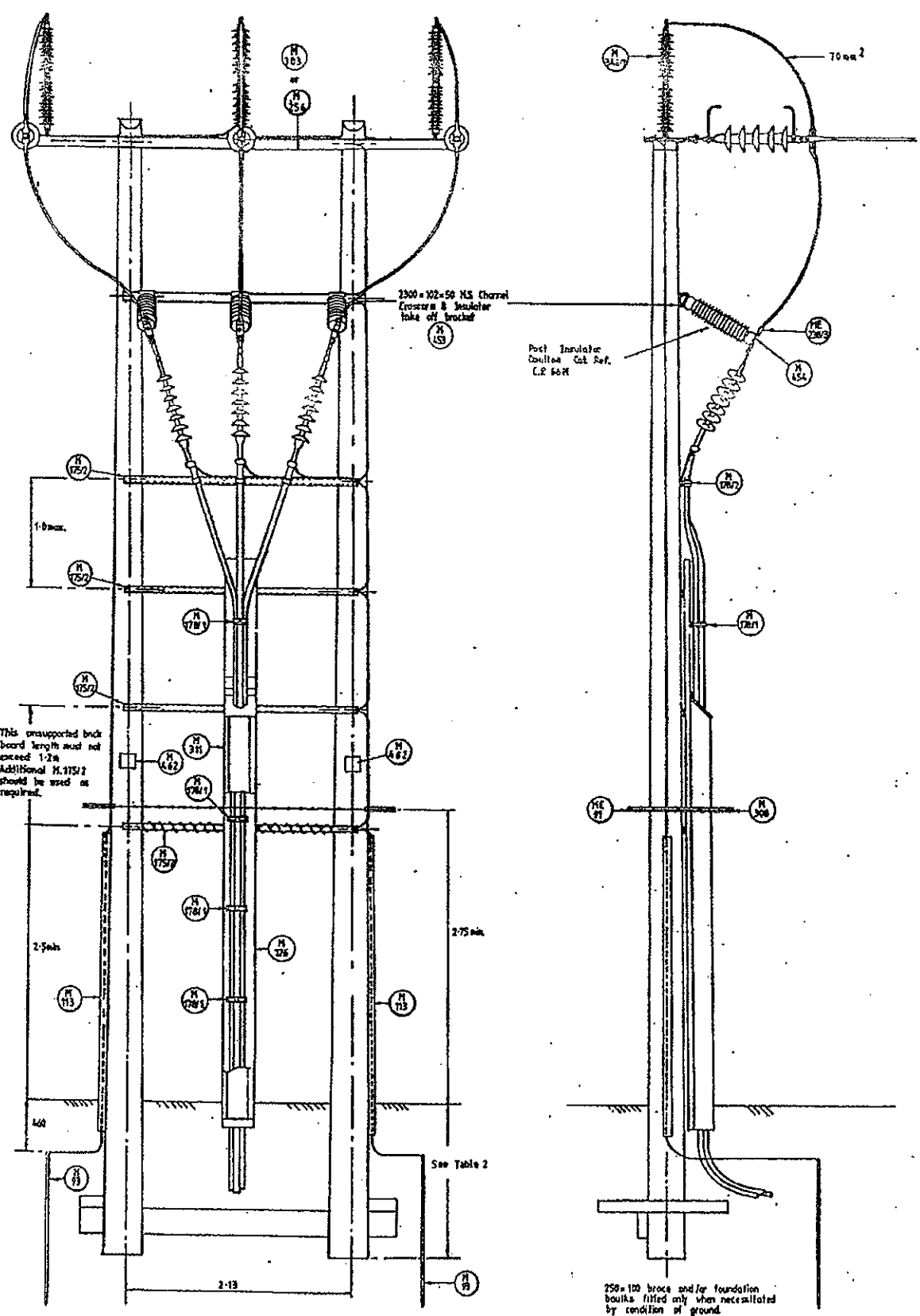
AS DRG MP 14/4

66KV. CONSTRUCTION
H' TERMINAL POLE

SCALE		 MEB Midlands Electricity plc
DATE	27.5.92	
DRAWN	W.B.	
TRACED	S.J.H.	
CHECKED		
APPROVED		 FIG 5

MATERIAL SCHEDULE FIG 5 (MF14/4)

ITEM	MIN	QUANTITY
Pole (size to be specified)	SPECIAL	2
Crossarm 200 AAAC Conductors	3451030	)
300 AAAC Conductors)		) 1
400 ACSR Conductors)	3452570	)
<u>Insulator Assembly 200 AAAC</u>		
Ball Ended Hook 70KN MFL	3450860	3
Disc Insulator Porcelain 70KN MFL	3450851	)
Glass 70KN MFL	3450852	) 15
Socket Tongue 70KN MFL	3450880	3
Arc Horn Live End	3450893	3
Dead End	3450894	3
<u>Insulator Assembly 300 AAAC + 400 ACSR</u>		
Shackle 125KN MFL	3452500	3
Ball Ended Eye Link 125KN MFL	3452490	3
Disc Insulator Porcelain 125KN MFL	3452541	15
300 AAAC Socket Tongue 125KN MFL	3452511	3
or 400 ACSR Socket Clevis 125KN MFL	3455100	3
+ Adapter 125KN MFL	3453601	3
Arc Horn Live End	3452552	3
Dead End	3452551	3
Conductor Terminating Clamp 200 AAAC 70KN	3455681	)
300 AAAC 125KN	3455684	) 3
400 ACSR	3452531	)
NOTE: FANDOWN MATERIALS SEE FIG 13		
Safety Sign Divs 3, 5 and 8	3454621	)
Divs 1, 4 and 7	3454622	) 2
Foundation Blocks (if required)		
1.25m x 250 x 100	3454411	2
3.2m x 250 x 100	3454424	1
Pole Bolt - Crossarm M20 x 260	3453633	2
Foundation Bolts (if required)		
M20 x 450	3453636	)
M20 x 530	3453637	) 4
M20 x 610	3453638	)
Square Flat Washer - M20 x 65	3453170	0-8*
Stay Materials/Stay		
Stay Wire - 19/3.55 - 700 Grade	3454562	AR
Stay Insulator	3454450	1
Dead End Stay Grip	3452703	3
Stay Thimble	3451152	3
Stay Rod	3450172	1
Stay Block	3454102	1
* With Foundation Blocks		



SCALE	
DATE	27.5.92
DRAWN	
TRACED	
CHECKED	
APPROVED	

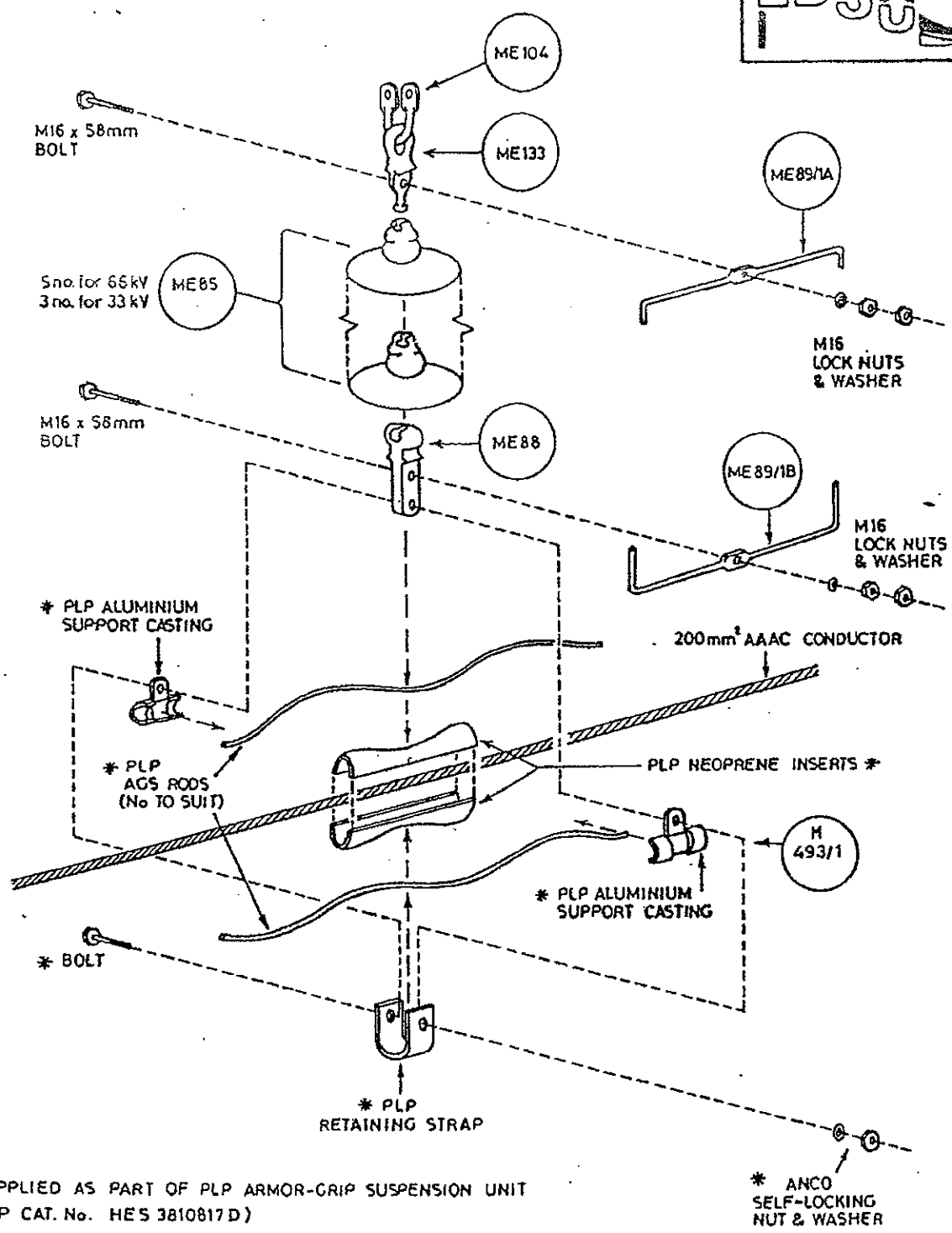
66 kV. CONSTRUCTION
CABLE TERMINAL POLES XLP CABLES

MEB
Midlands Electricity plc

FIG. 6

MATERIAL SCHEDULE FIG 6

ITEM	MIN	QUANTITY
NOTE: For Pole, Terminal Crossarm, Insulator Assembly Foundation Blocks refer to Fig 5 Material List.		
Post Insulator Support Crossarm	3454530	1
Cable Support Crossarm	3451752	AR
Cable Back Board	3451760	AR
Cable Trunking	3453110	AR
Cable Cleats	3451781	AR
Post Insulator	345	3
Cable Connector Bar	3454540	3
Compression Dead End 200 AAAC	3455667	)
300 AAAC	3455669	) 3
400 ACSR	3452534	)
Earth Conductor	345	
Earth Rod	345	
Earth Conductor Cover Board	345	

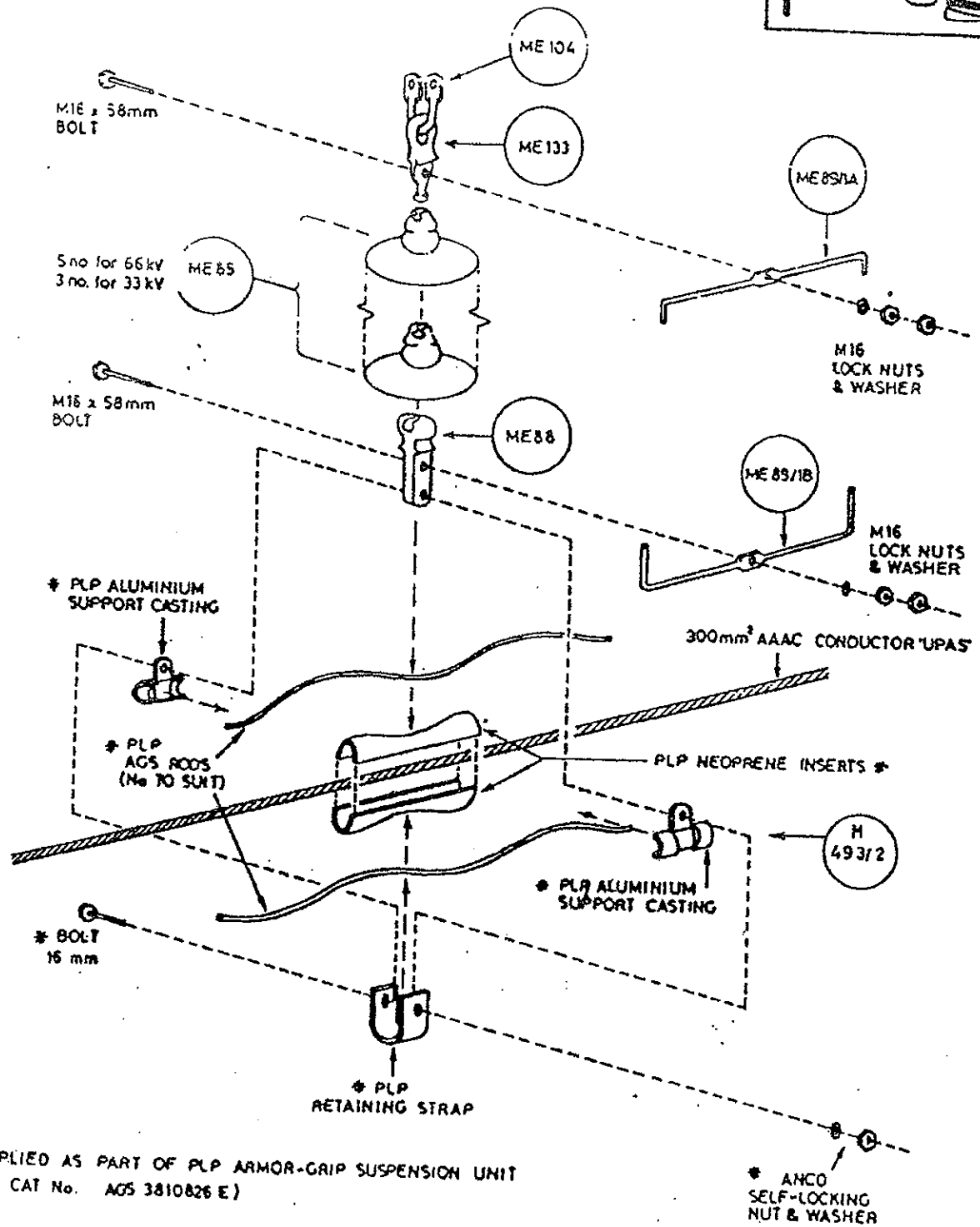


AMCO 15828.90 AMCO BY DATE	TITLE SUSPENSION INSULATOR ASSEMBLY 200 AAAC CONDUCTORS	SCALE	—
		DATE	22.5.92
		DRAWN	
		CHECKED	
		APPROVED	

MEB

Midlands Electricity plc

FIG.7



TITLE

SUSPENSION INSULATOR ASSEMBLY
300 AAAC CONDUCTORS

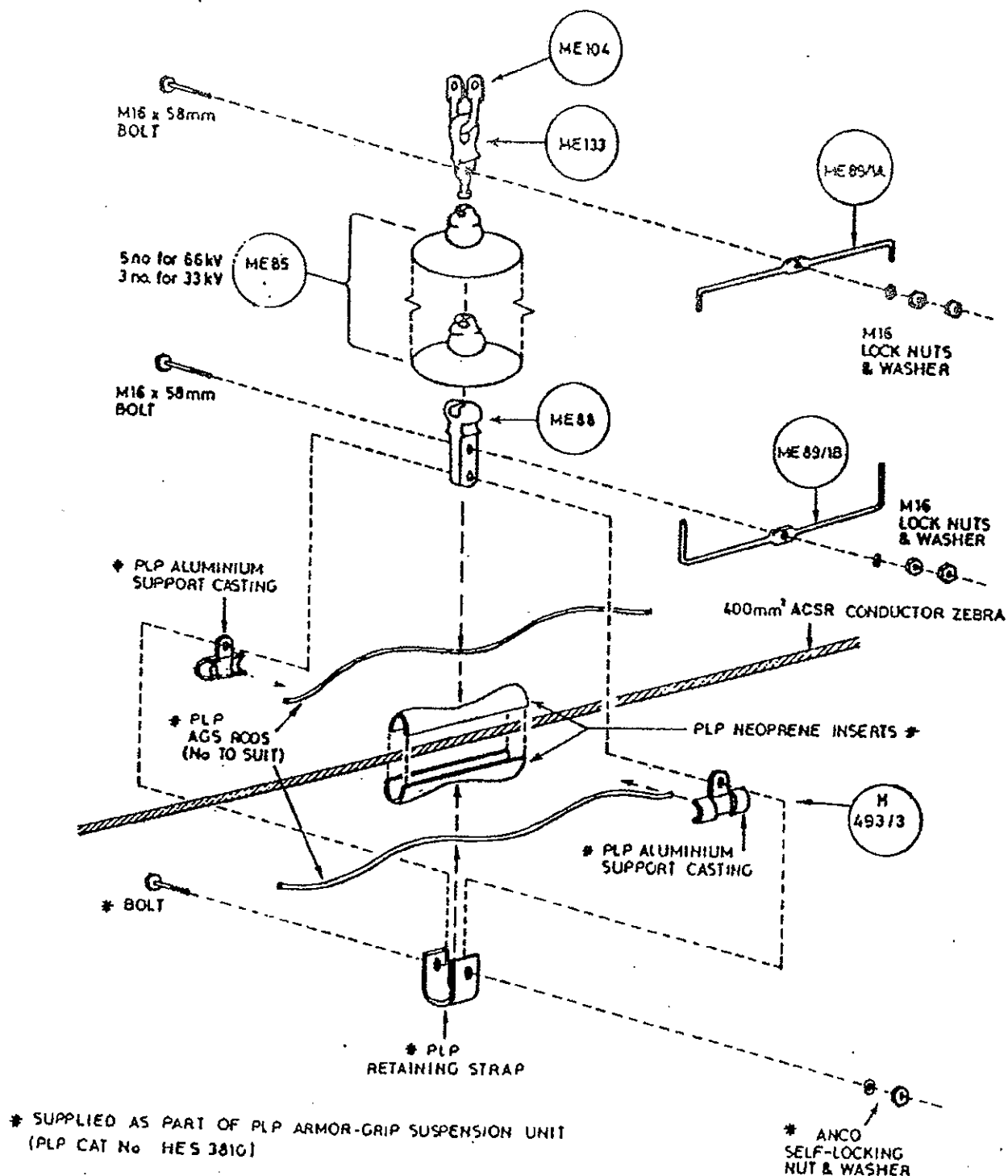
SCALE	
DATE	22-5-92
DRAWN	
TRACED	
CHECKED	
APPROVED	

MEB
Midlands Electricity plc

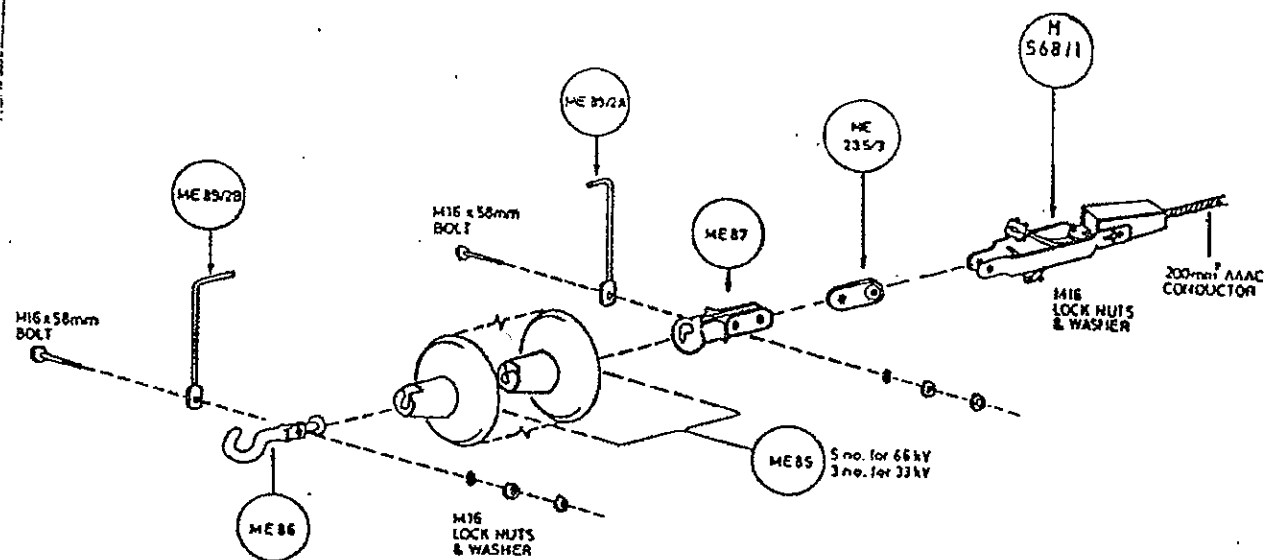
FIG 8

APPROVED BY
DATE

AMDO 15828.90



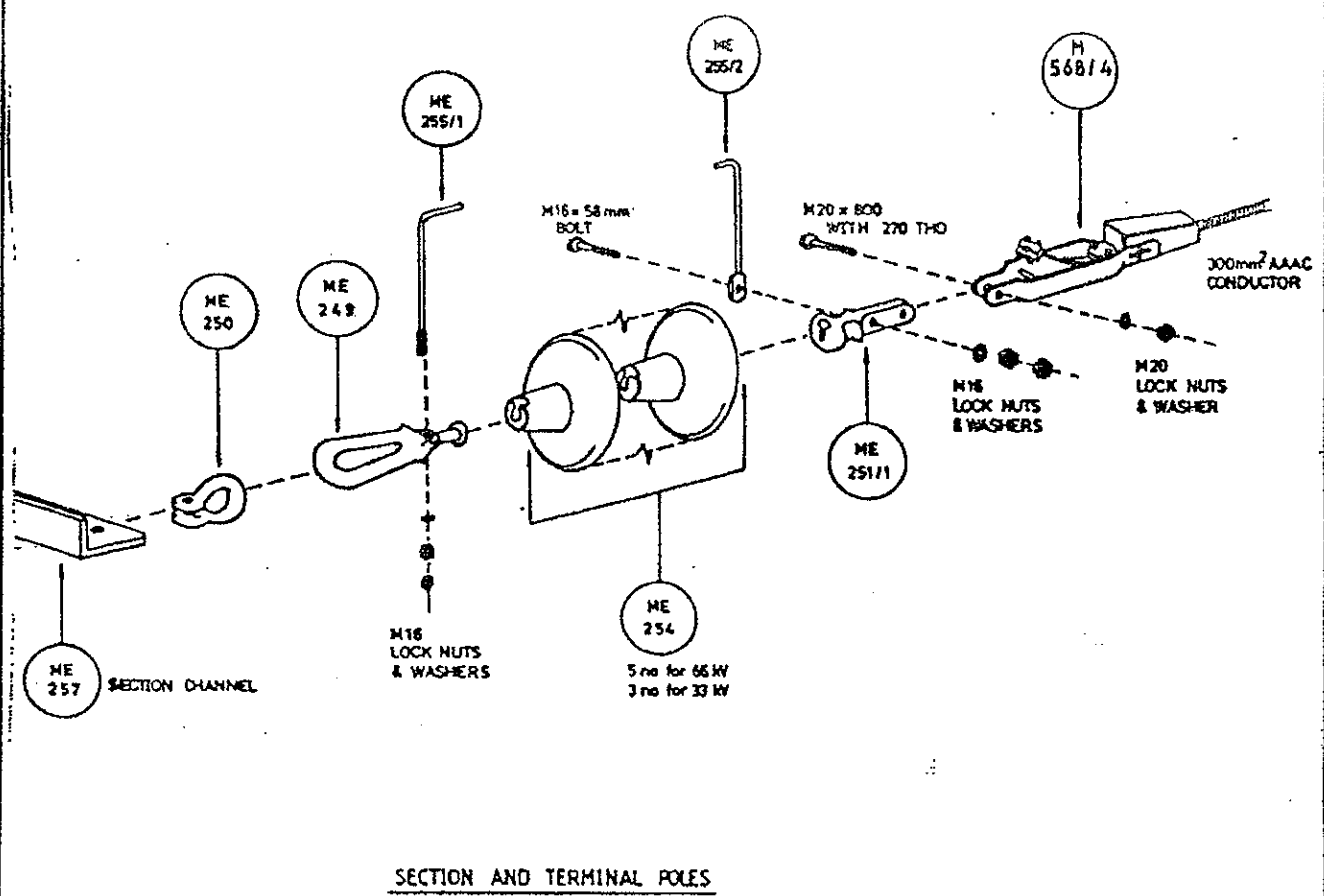
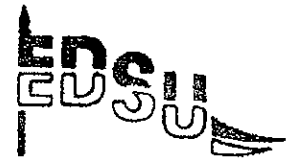
Approved by DATE	TITLE SUSPENSION INSULATOR ASSEMBLY 400 ACSR CONDUCTORS	SCALE	—	MEB Midlands Electricity plc DWG No FIG. 9
		DATE	22-5-92	
		DRAWN		
		TRACED		
		CHECKED		
		APPROVED		



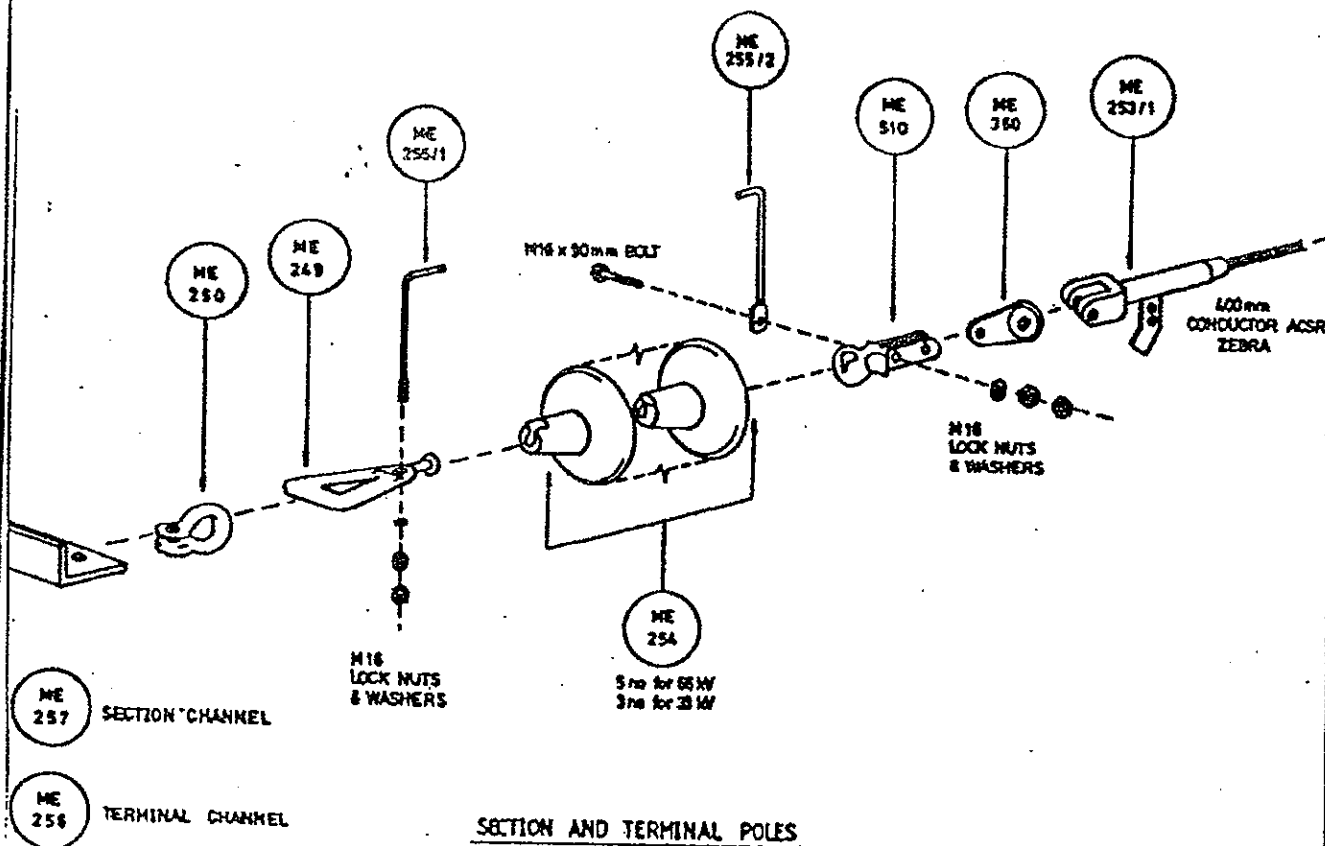
SECTION AND TERMINAL POLES

ADD'D BY DATE		TITLE TENSION INSULATOR ASSEMBLY 200 AAAC CONDUCTORS	SCALE	—	MEB Midlands Electricity plc ONE No FIG. 10
			DATE	22-5-92	
			DRAWN		
			TRACED		
			CHECKED		
			APPROVED		

AMDC 15628.90



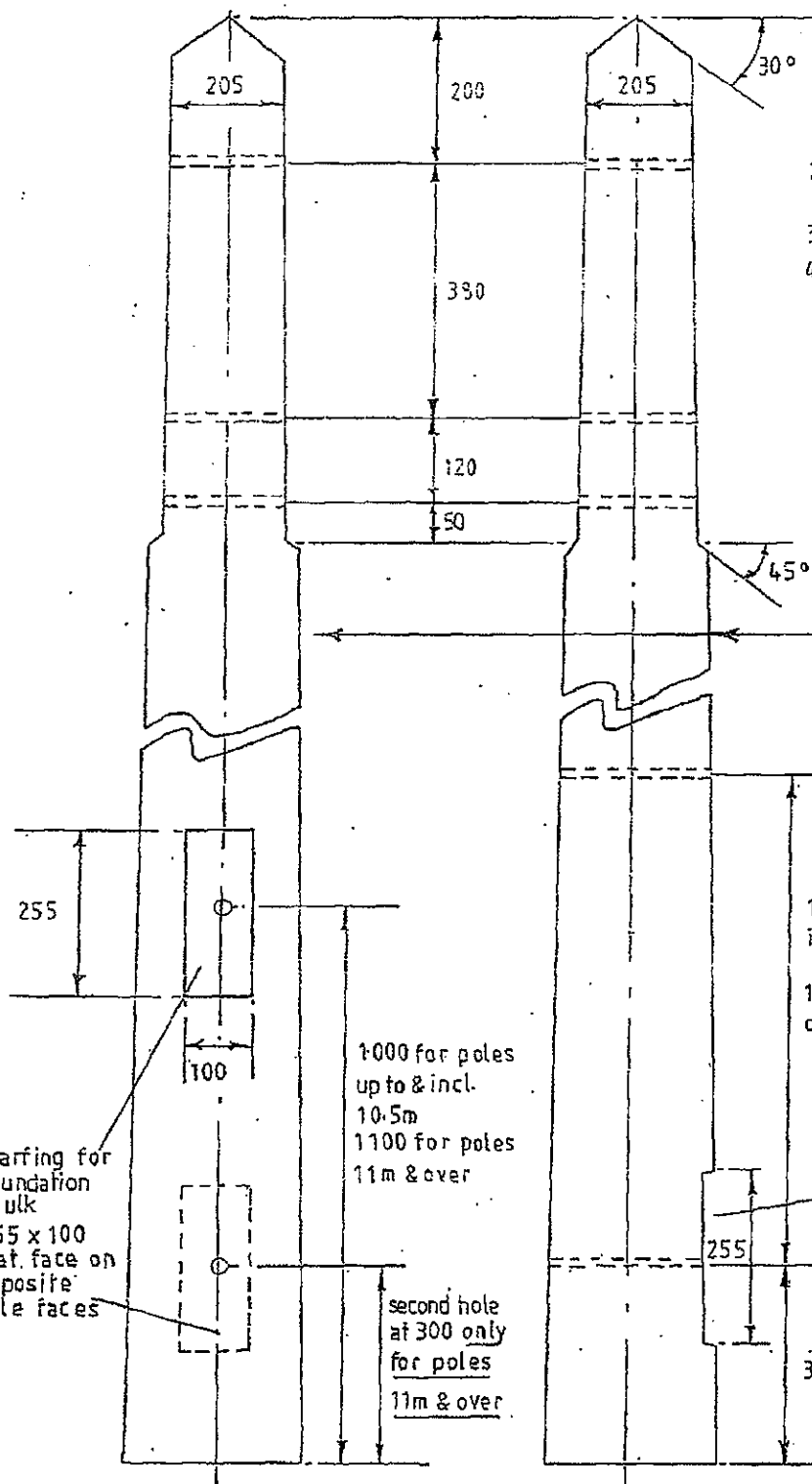
1710 IN DRAWING	TITLE TENSION INSULATOR ASSEMBLY 300 AAAC CONDUCTORS	SCALE	—	MEB Midlands Electricity plc DWG No FIG. 11
		DATE	22-5-92	
		DRAWN		
		TRACED		
		CHECKED		
		APPROVED		



AMDO 15628.90 AMDO BY DATE	TITLE TENSION INSULATOR ASSEMBLY 400 ACSR CONDUCTORS	SCALE	—	MEB Midlands Electricity plc FIG. 12
		DATE	22-6-92	
		DRAWN	—	
		TRACED		
		CHECKED		
		APPROVED		

SINGLE
POLES

MATCHED
H POLES



1. ALL DRILLINGS TO BE 22 mm ϕ
2. POLES TREATED TO CURRENT EATS 43-88 PARTS 1&2
3. DIMENSIONS IN MM
4. POLES SCARFED ON BOTH SIDES TO GIVE 205 BETWEEN FACES

GOUGE MARK AND POLE INFORMATION ON THIS FACE OF POLE

1220 for poles up to and including 14.0m

1520 for poles 15.0m and over

1000 for poles up to & incl. 10.5m
1100 for poles 11m & over

second hole at 300 only for poles 11m & over

Scarfing to give 255 x 100 mm flat face

Scarfing for foundation baulk 255 x 100 flat face on opposite pole faces

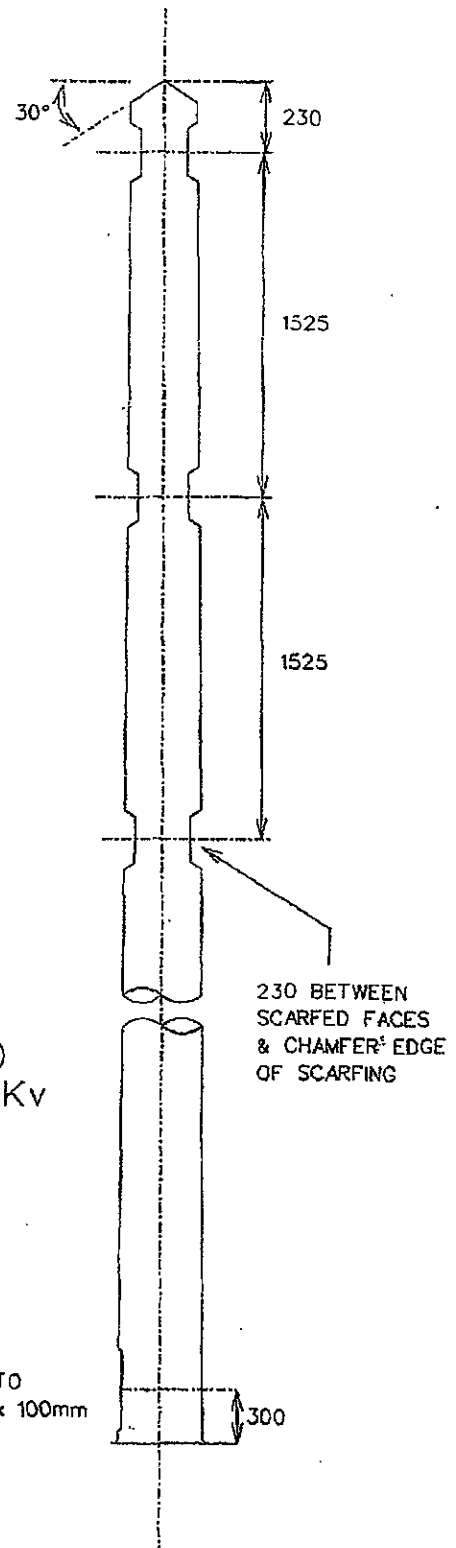
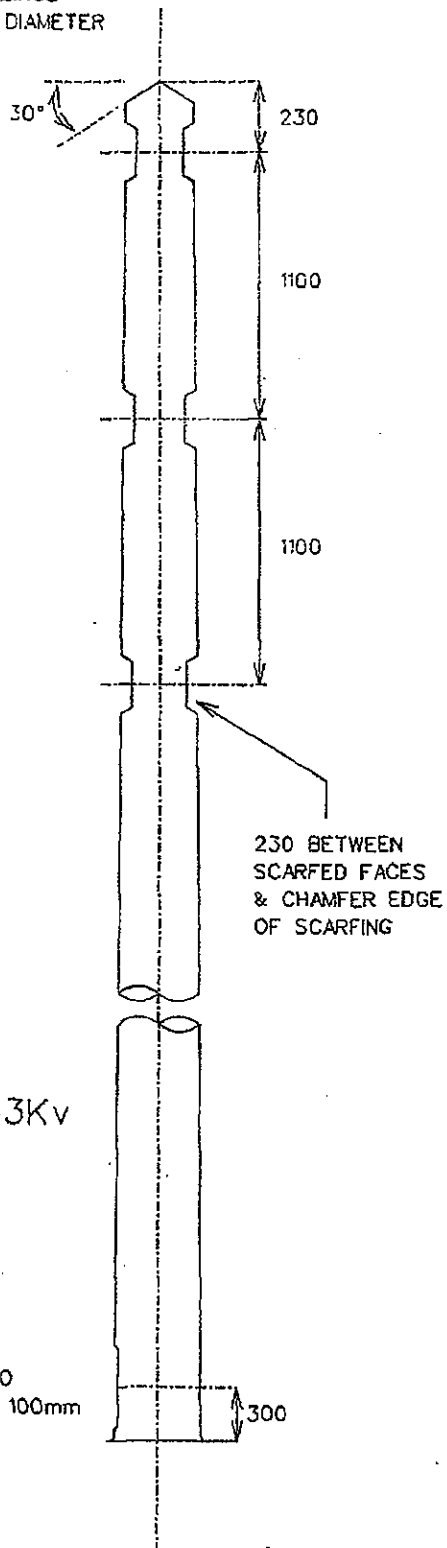
AM0015628E
B
DATE 7.8.96
ISSUE D AUG.96
- REPLACES
ISSUE 'A'

TITLE
STANDARD POLE DRILLINGS
AND SCARFING
STOUT CLASS POLES
(INCL. EXTRA STOUT)

SCALE	—
DATE	6.8.96
DRAWN	G.P.
TRACED	E.L.W.
CHECKED	
APPROVED	

MEB
Midlands Electricity plc
DRG No.
M 450/3

ALL POLE DRILLINGS
TO BE 22mm DIAMETER



ALL DIMENSIONS IN mm EXCEPT WHERE OTHER WISE STATED

SCALE	-
DATE	22.01.03
DRAWN	MJB
TRACED	-
CHECKED	-
APPROVED	-

DUAL CIRCUIT POLE
DRILLING & SCARFINGS

ISSUE DATE

Produced by
Primary CAD



Aquila

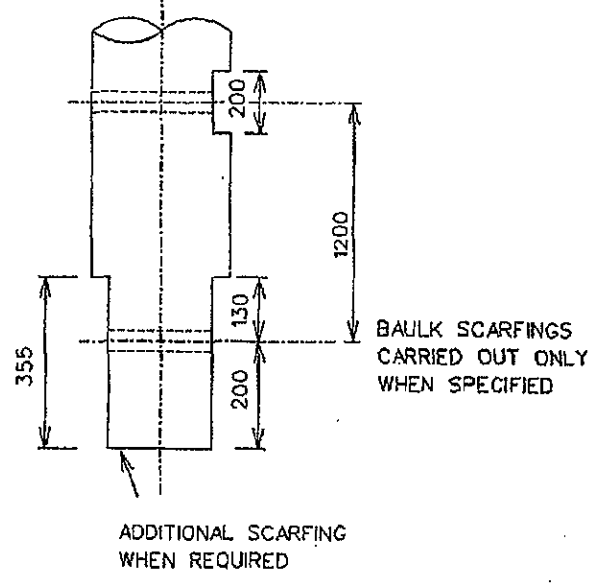
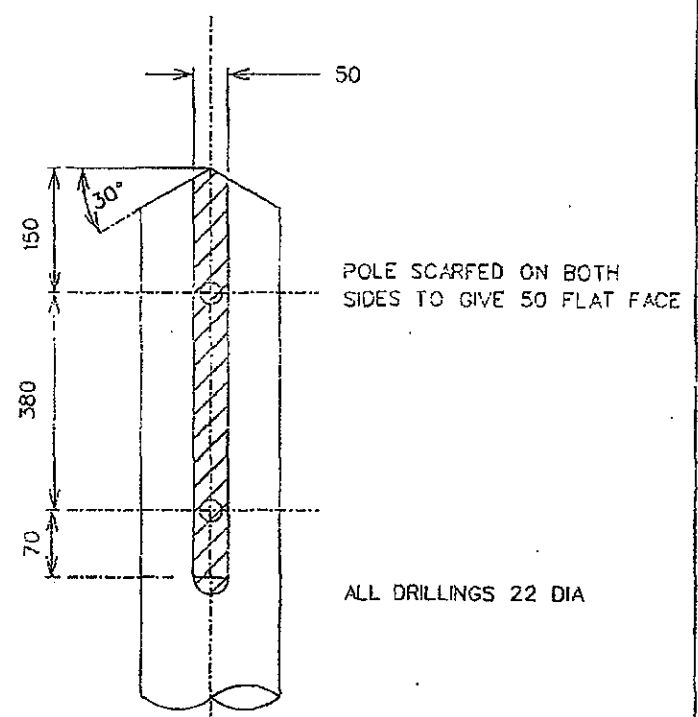
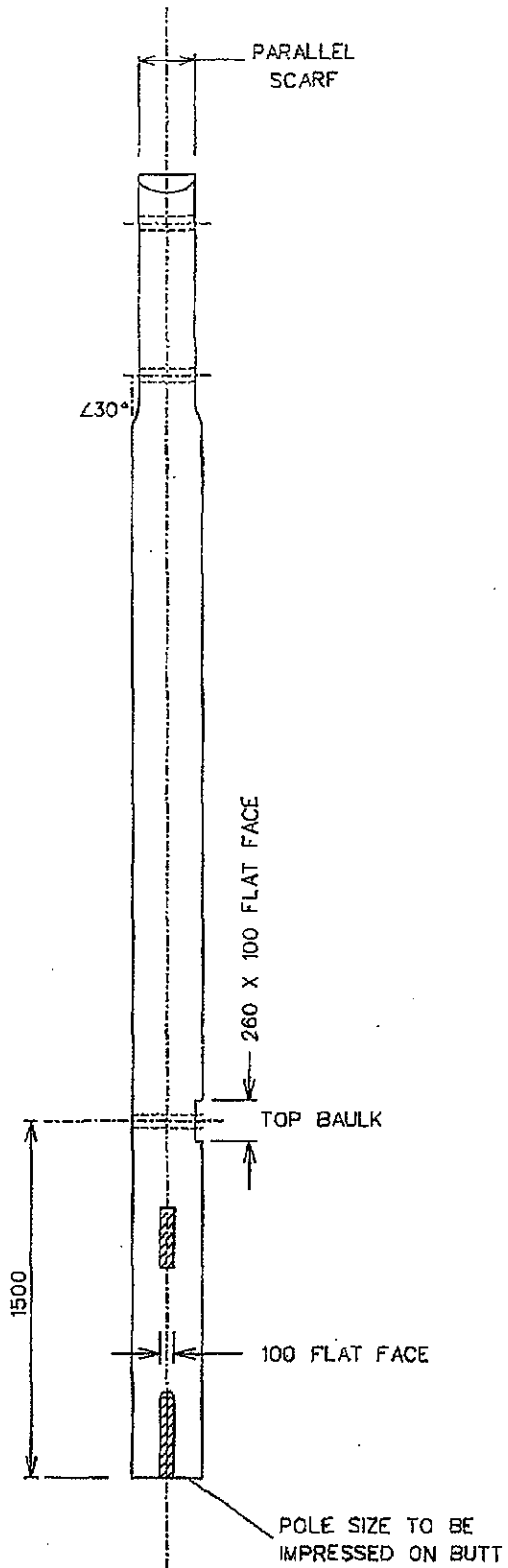
DRG. No.

M450-4

DRG. No.

M450-4

A4



ALL DIMENSIONS ARE IN mm EXCEPT WHERE OTHERWISE STATED

SCALE	-
DATE	20.01.03
DRAWN	MJB
TRACED	-
CHECKED	-
APPROVED	-

TRIDENT H-POLE DRILLINGS & SCARFINGS INTERMEDIATE SUPPORTS

ISSUE DATE

Produced by
Output Services



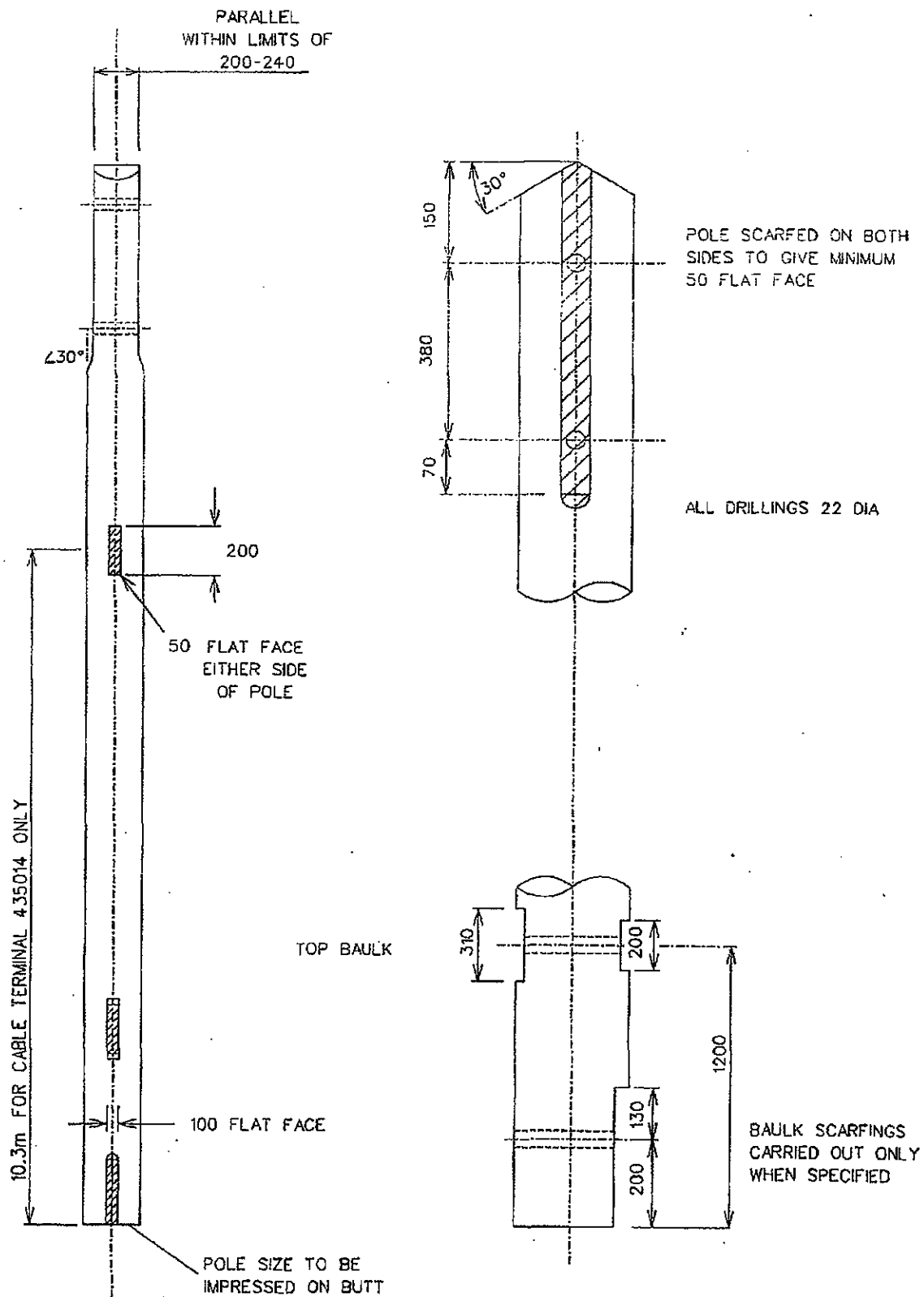
Aquila

DRG. No.

M450-5

DRG. No.

M450-5



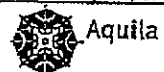
ALL DIMENSIONS ARE IN mm EXCEPT WHERE OTHERWISE STATED

SCALE	-
DATE	20.01.03
DRAWN	MJB
TRACED	-
CHECKED	-
APPROVED	-

TRIDENT H-POLE DRILLINGS & SCARFINGS
INTERMEDIATE SUPPORTS
TERMINAL + SECTION SUPPORTS

ISSUE DATE

Produced by
Output Services



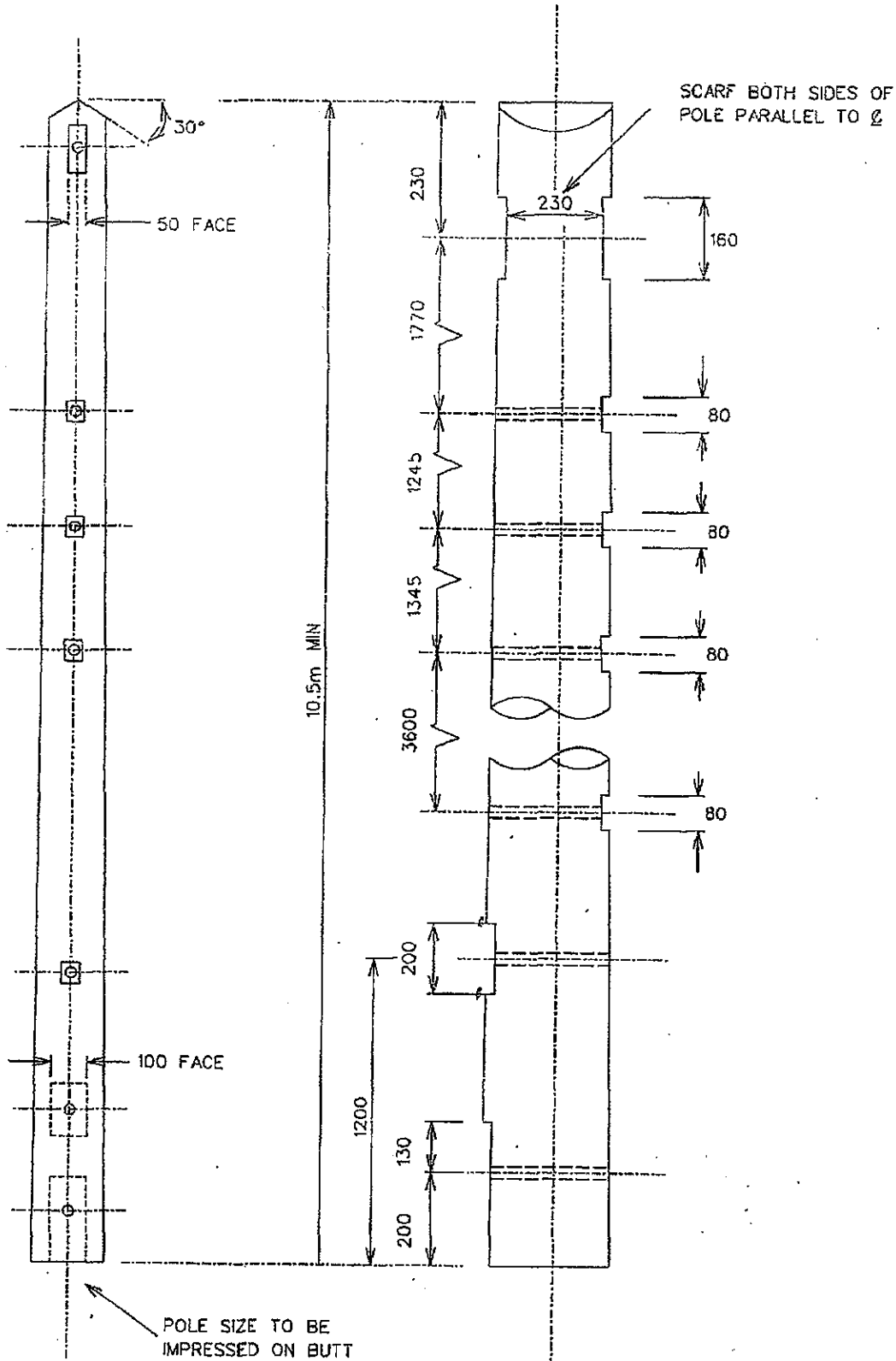
DRG. No.

M450-6

DRG. No.

M450-6

A1



ALL DIMENSIONS 22 DIA ALL DIMENSIONS IN mm EXCEPT WHERE OTHER WISE STATED

SCALE	-
DATE	22.01.03
DRAWN	MJB
TRACED	-
CHECKED	-
APPROVED	-

TRIDENT AUXILIARY POLE
DRILLINGS & SCARFINGS

ISSUE	DATE

Produced by
Output Services



Aquila

DRG. No.

M450-7

DRG. No.

M450-7